



Swiss Institute of
Bioinformatics



University of
Zurich^{UZH}

ELIXIR Spatial Omics Data Analysis

08 Oct 2026

(Spatial) Statistics for spatial omics data

Peiying Cai

Mark D. Robinson lab

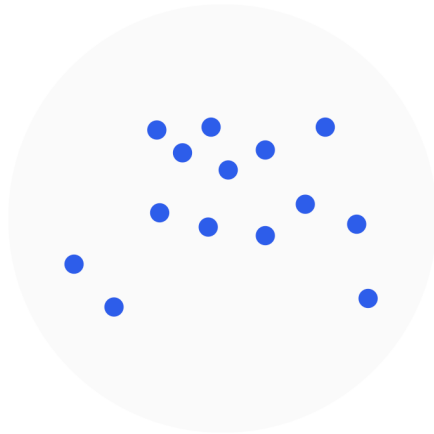
Statistical Bioinformatics Group, DMLS@UZH+SIB

<https://robinsonlabuzh.github.io/>



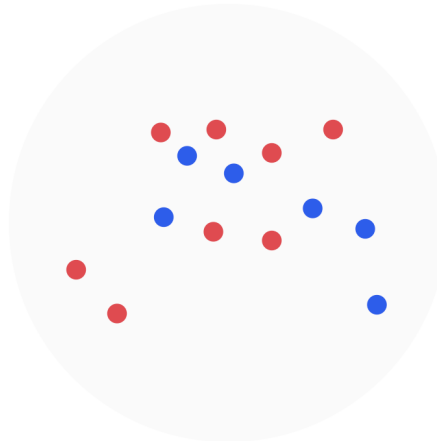
Why spatial statistics?

Are these cells clustered?



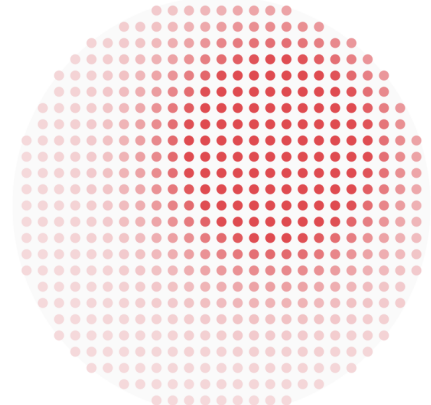
Point pattern

Do two cell types co-localise?



Marked point pattern

Is gene expression spatially autocorrelated?



Values at fixed spots

- Spatial organisation can itself be biological signal.
- Tobler's "first law of geography": everything is related to everything else, but near things are more related than distant things.

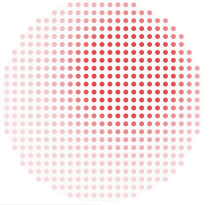


What is random? Two views of spatial data



Event-based view

- Locations are treated as stochastic
- Point pattern is the outcome
- Point-pattern analysis

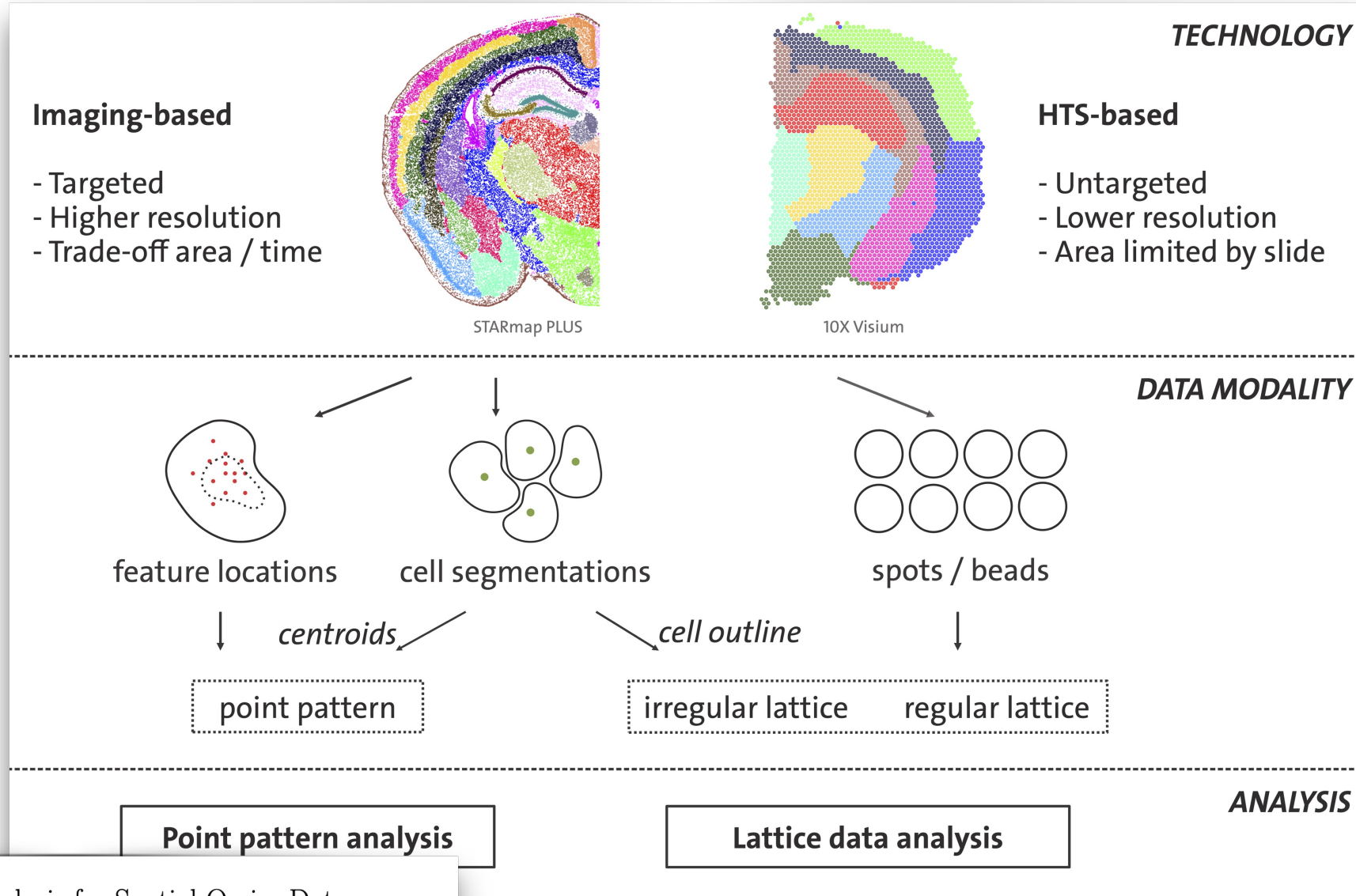


Observation-based view

- Locations are treated as fixed
- Values at locations are random variables
- Lattice analysis



Main concept: different data representations



Martin



Samuel

pasta: Pattern Analysis for Spatial Omics Data

Martin Emons^{1,†}, Samuel Gunz^{1,†}, Helena L. Crowell², Izaskun Mallona¹, Reinhard Furrer³, and Mark D. Robinson^{1,*}

Emons et al., Nucleic Acids Res. 2025



Data representations determine spatial statistics options

C

ANALYSIS

Point pattern

Lattice data

Quantification of spatial arrangement
(e.g., spacing vs. clustering).

Quantification of measurements in a defined
neighborhood and their association.

Categorical (e.g., cell types)

Uni-
variate

Colocalization of one cell type and
at which scale?

K, L and G functions

Bi-
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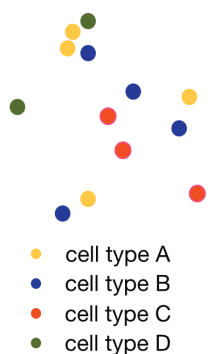
Colocalization between two cell
types and at which scale?

Cross K, L and G functions

Multi-
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Colocalization of one cell
to a set of other cell types?

Dot functions



not common

How often are spots of the same
cluster neighbouring each other?

Join count statistics

Which clusters are found more
frequently neighbouring each other?

Multivariate join count statistics



Numerical (e.g., gene expression)

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At which scale is there (spatial)
correlation of gene expression?

Mark correlation function

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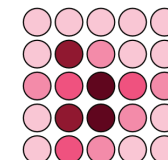
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Spatial autocorrelation of a gene?
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Spatial correlation of two genes?
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Spatial correlation of a set of genes?
Multivariate Geary's C



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Point patterns



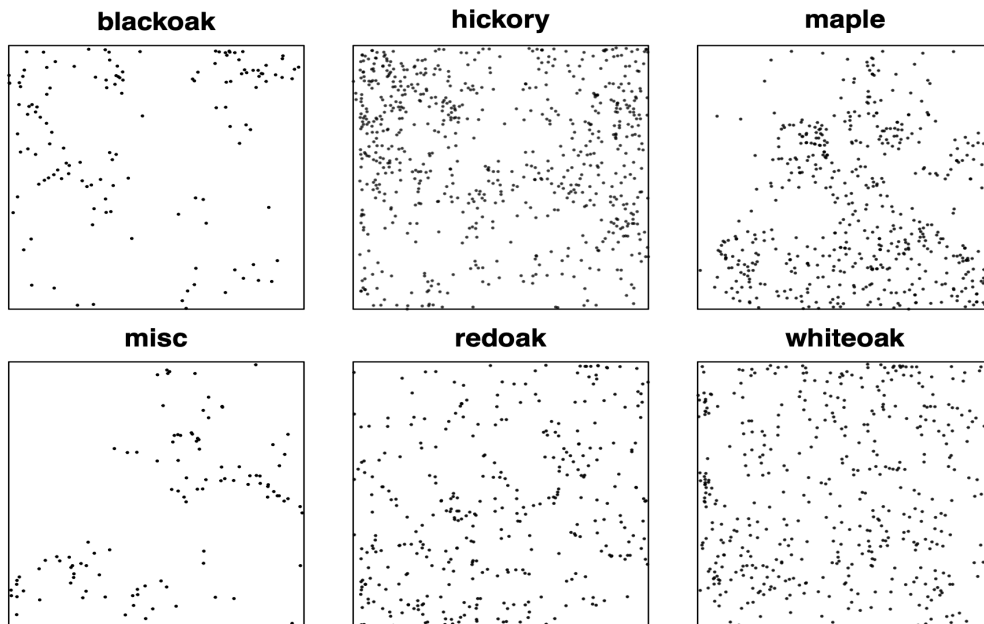
What are point patterns?

- **Locations** themselves are the **observations** of interest
- Often represented as a realisation of a **spatial point process**
- Points may carry **marks**, e.g., cell types (discrete) or gene expression (continuous)
- Here we mostly focus on discrete categories, i.e., patterns within and between cell types



What are point patterns?

“a realisation of a spatial point process effectively assumes that the locations of points are not fixed, and that the point pattern is the response or observation of interest.”

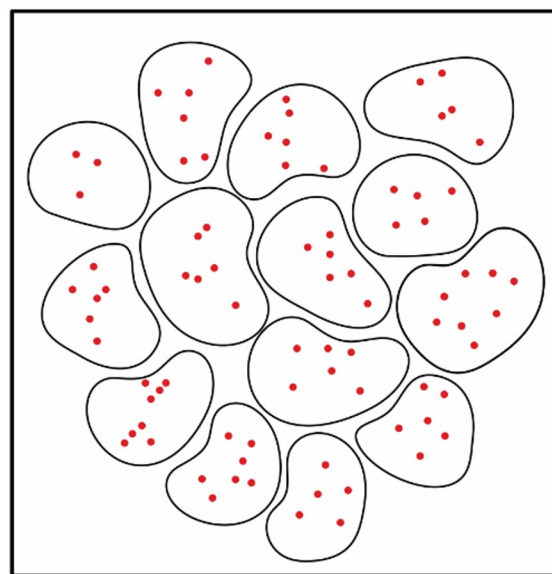


- ❖ Location of trees according to their botanical classification
- ❖ Cholera cases in a city
- ❖ Epicentres of earthquakes along a fault line

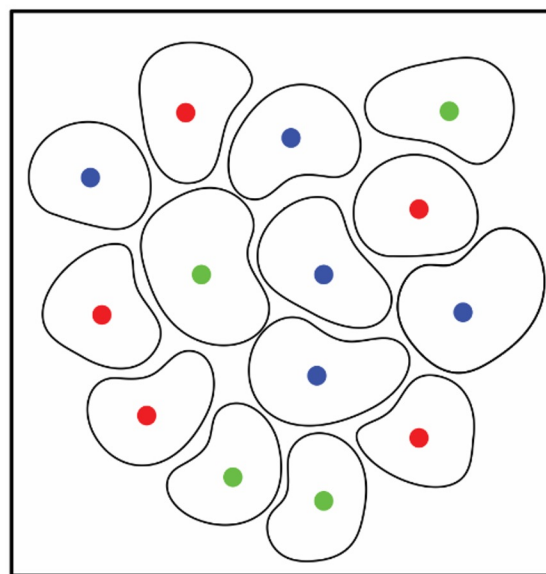


Approximating cells as points

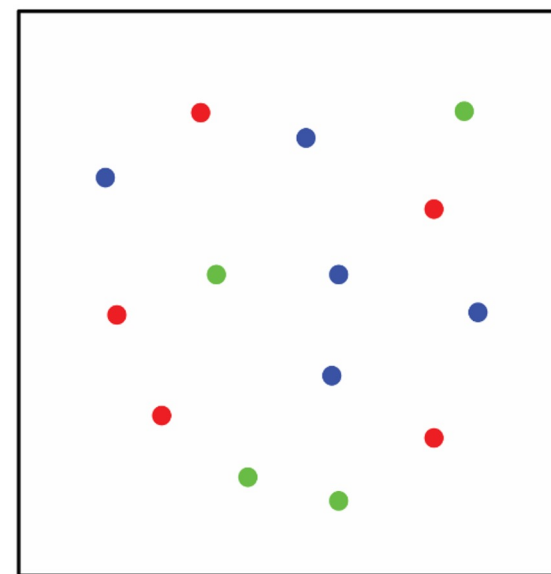
A) Point pattern approximation of cells



Cells with molecular features



Centroids with cell type labels



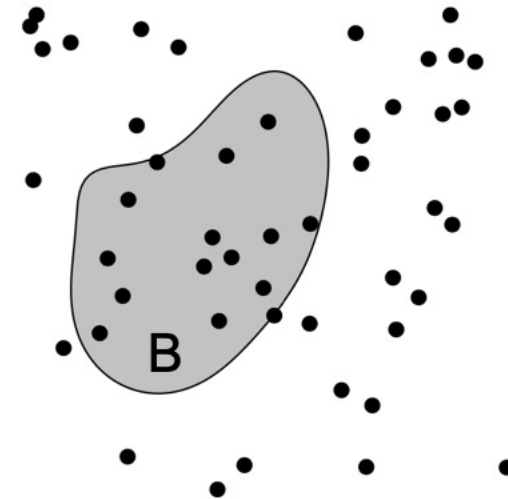
Marked point pattern



First-order property: intensity

- $\mathbf{X}$ is the point process; x is the (observed) point pattern
- λ : intensity function, the mean number of events per unit area

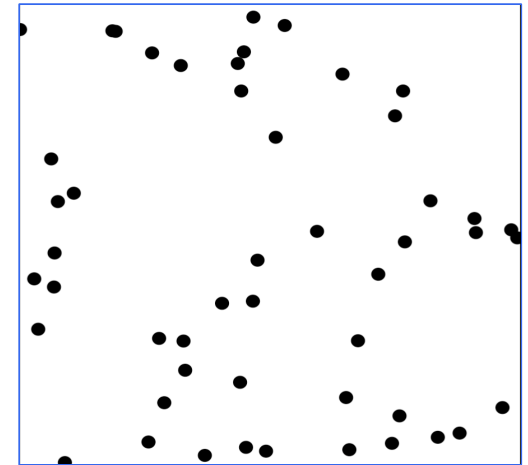
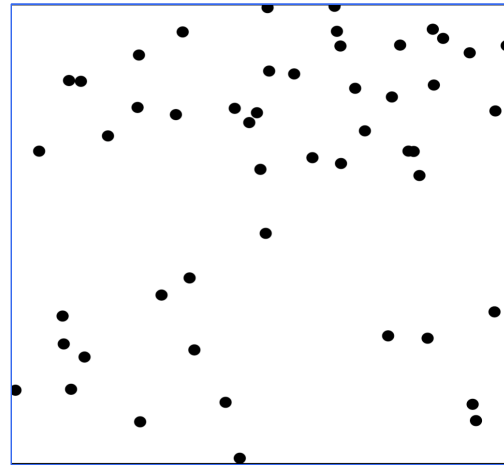
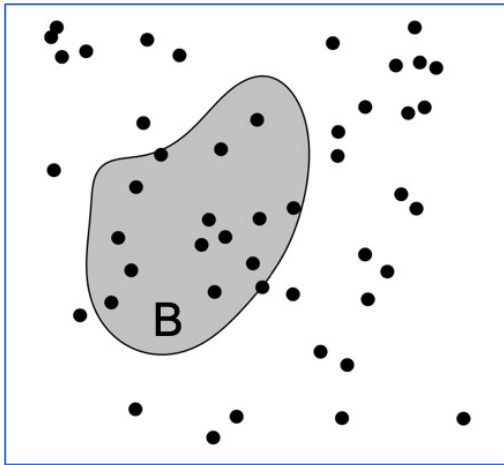
$$\mathbb{E}[n(\mathbf{X} \cap B)] = \int_B \lambda(u) \, du$$





Homogeneous intensity

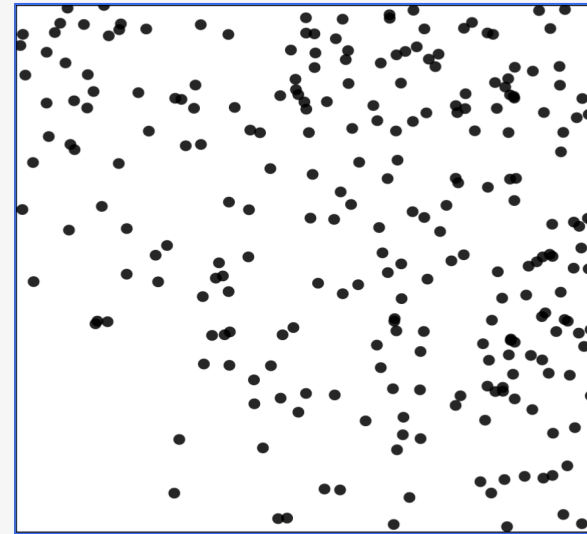
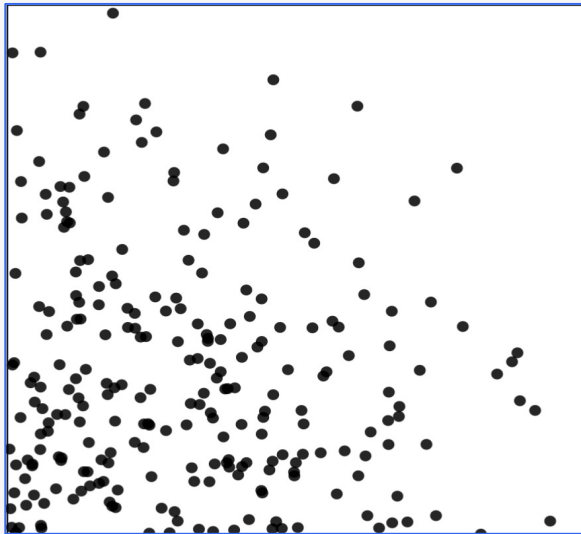
- The points have no preference for any spatial location.
- The number $n(X \cap B)$ of random points falling in a test region B has mean value $E[n(X \cap B)] = \lambda|B|$.





Inhomogeneous intensity

Intensity varies with location: $\lambda(u)$ is not constant.





Independence and complete spatial randomness

Independence

- Information about the outcome in one region of space has no influence on the outcome in other regions;
- For test regions $B_1, B_2, \dots, B_m$ which do not overlap, the counts $n(X \cap B_1), \dots, n(X \cap B_m)$ are independent random variables.



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Complete spatial randomness (CSR): homogeneous & independent



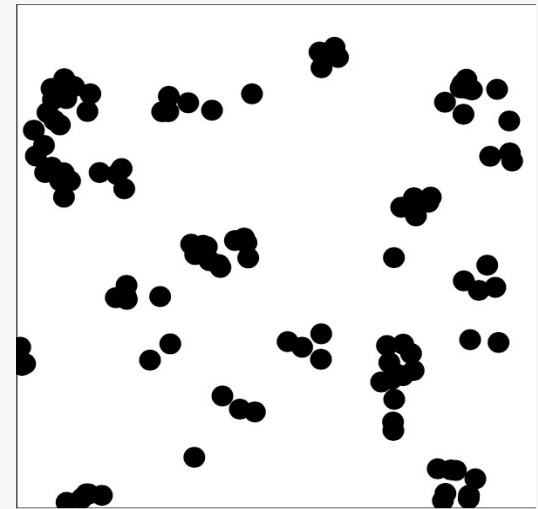
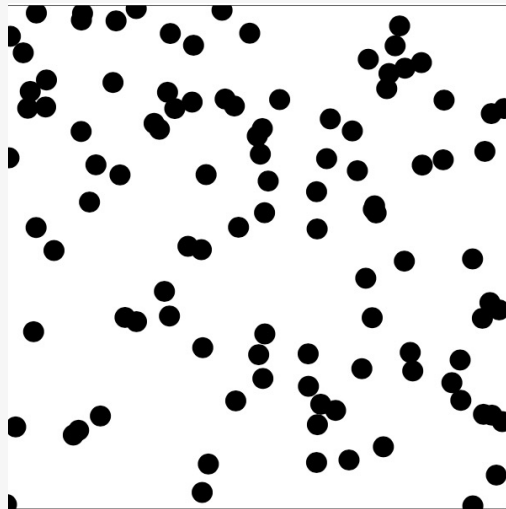
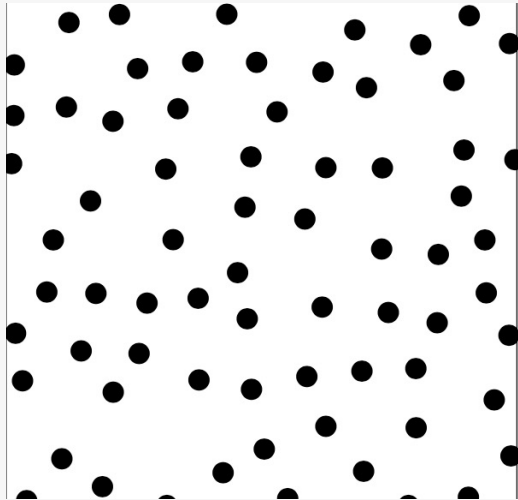
Properties of point patterns

Which of these is homogeneous?

Which of these is completely spatially random (CSR)?

Which of these is clustered?

Which of these is not independent?



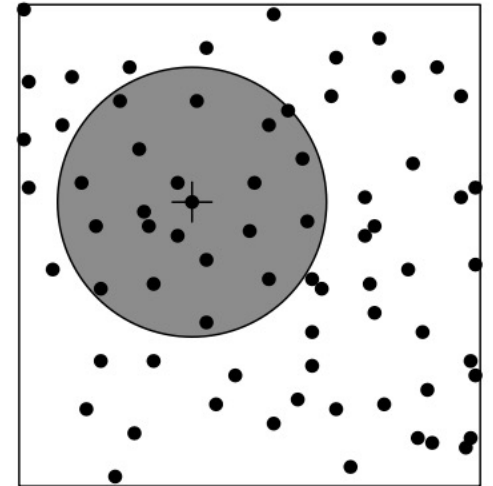


Second-order: correlation

Ripley's K function:

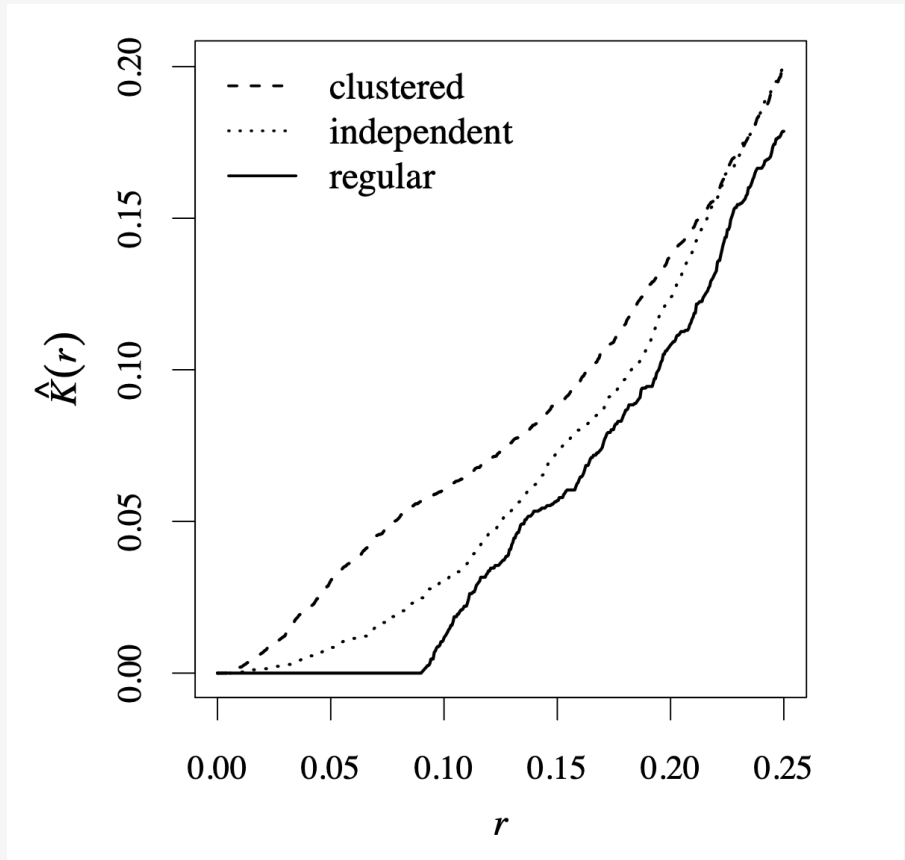
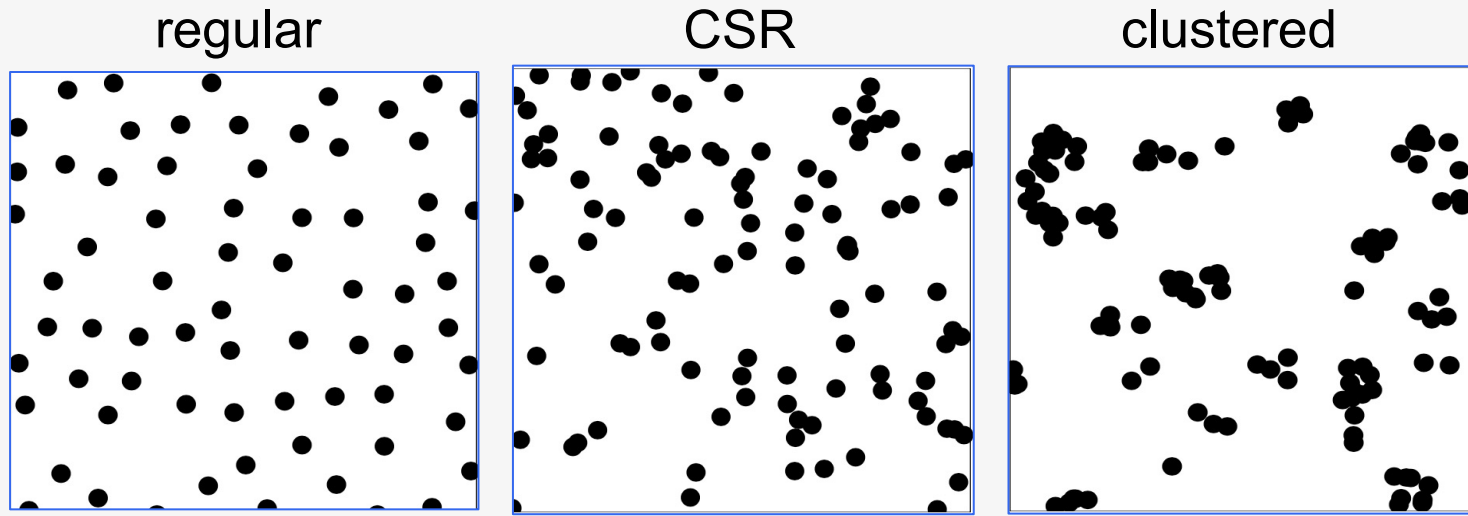
"the empirical K-function $K(r)$ is the cumulative average number of data points lying within a distance r of a typical data point."

$$K(r) = \frac{1}{\lambda} \mathbb{E} [\text{number of } r\text{-neighbours of } u \mid \mathbf{X} \text{ has a point at location } u]$$





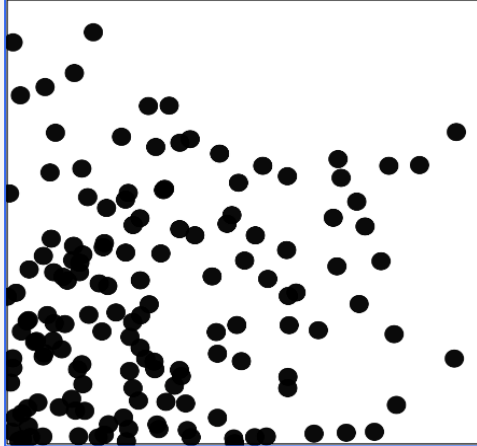
Ripley's K distinguishes spatial interaction



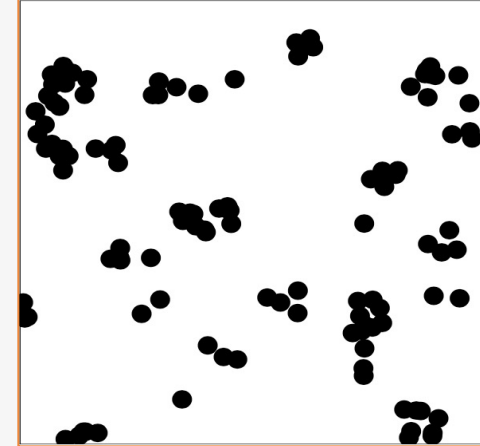


Is it clustering or intensity?

Inhomogeneous intensity



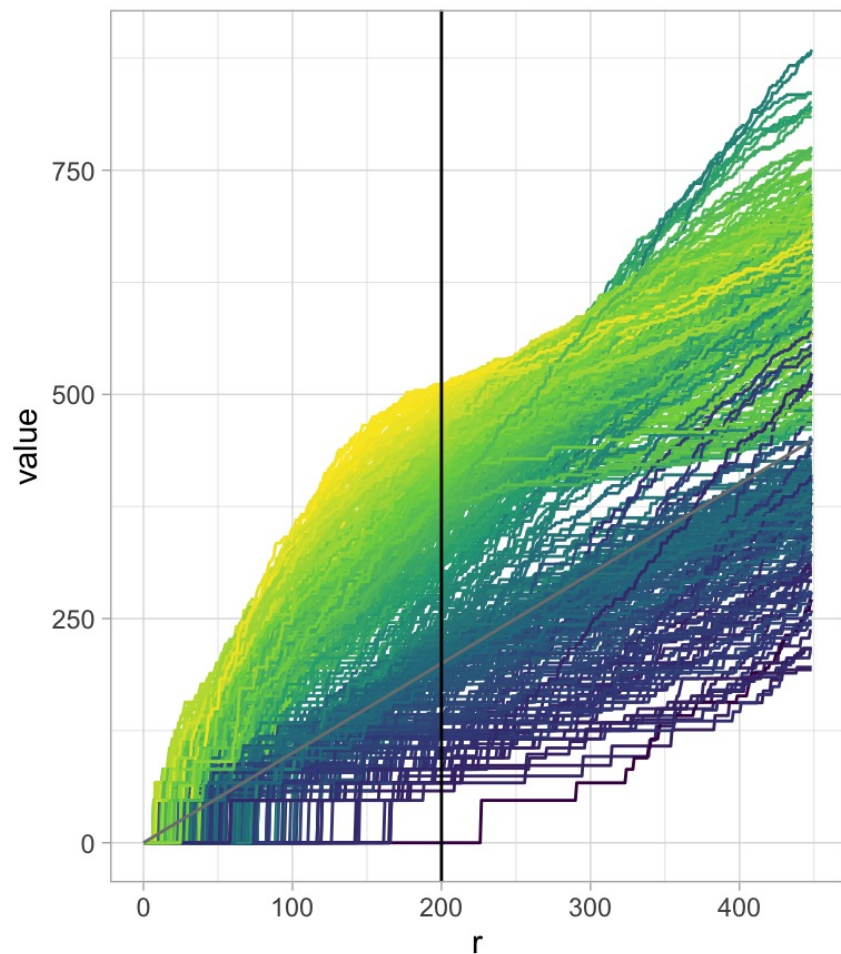
Genuine interaction
between points (clustering)



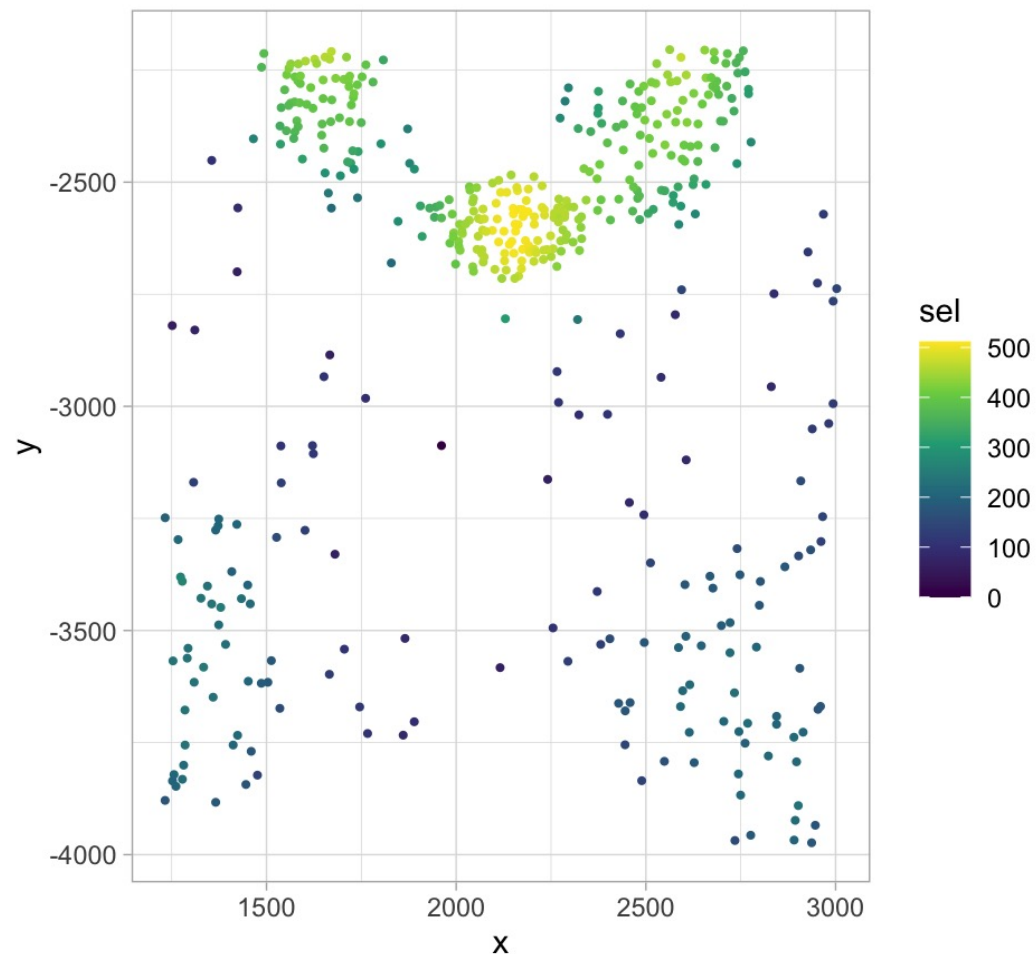


LISA: “local” indicators of spatial association

LISA curves of slice 0.01

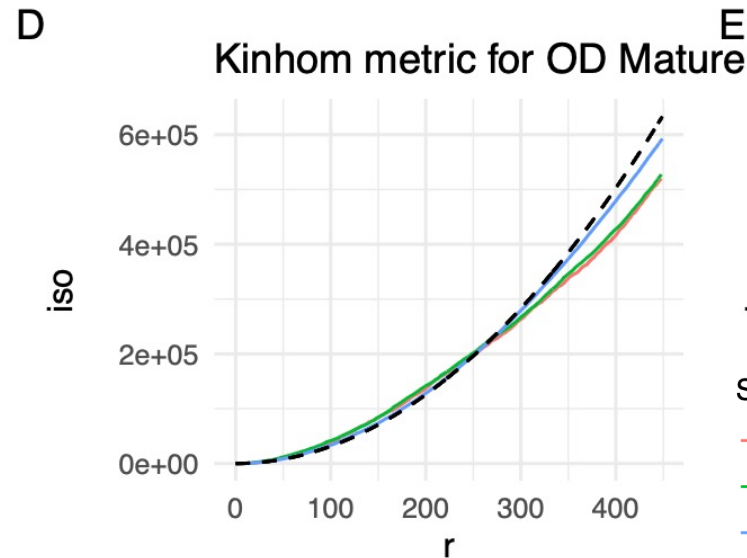
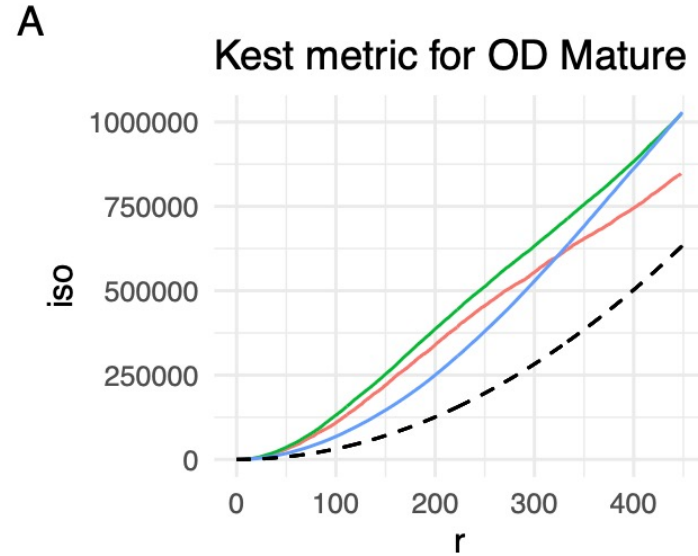
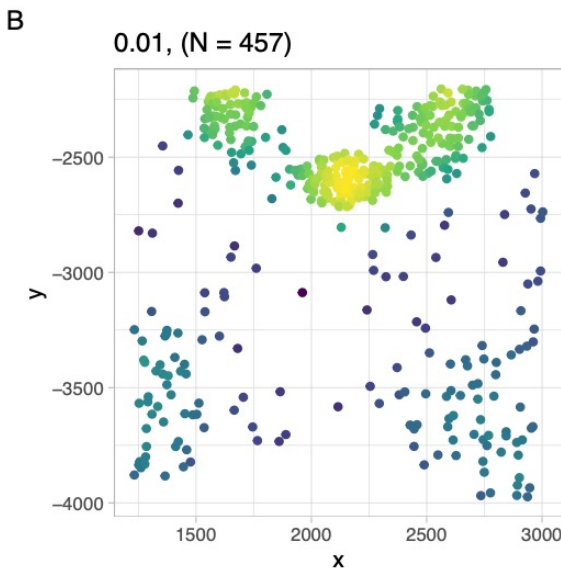
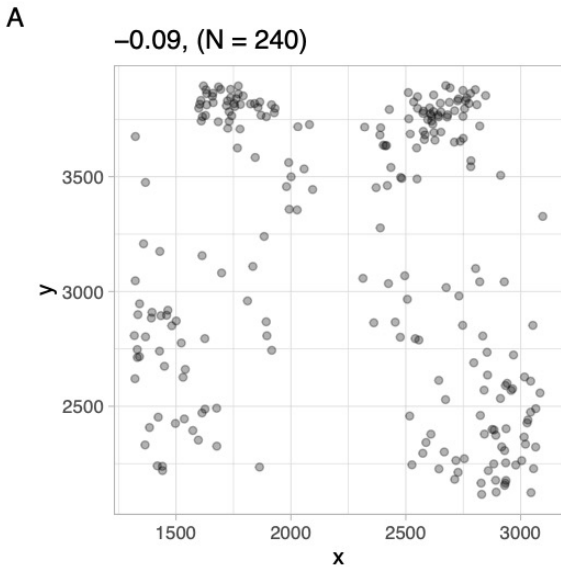


Points coloured by LISA value at r ~ 200





Confounding between clustering and intensity

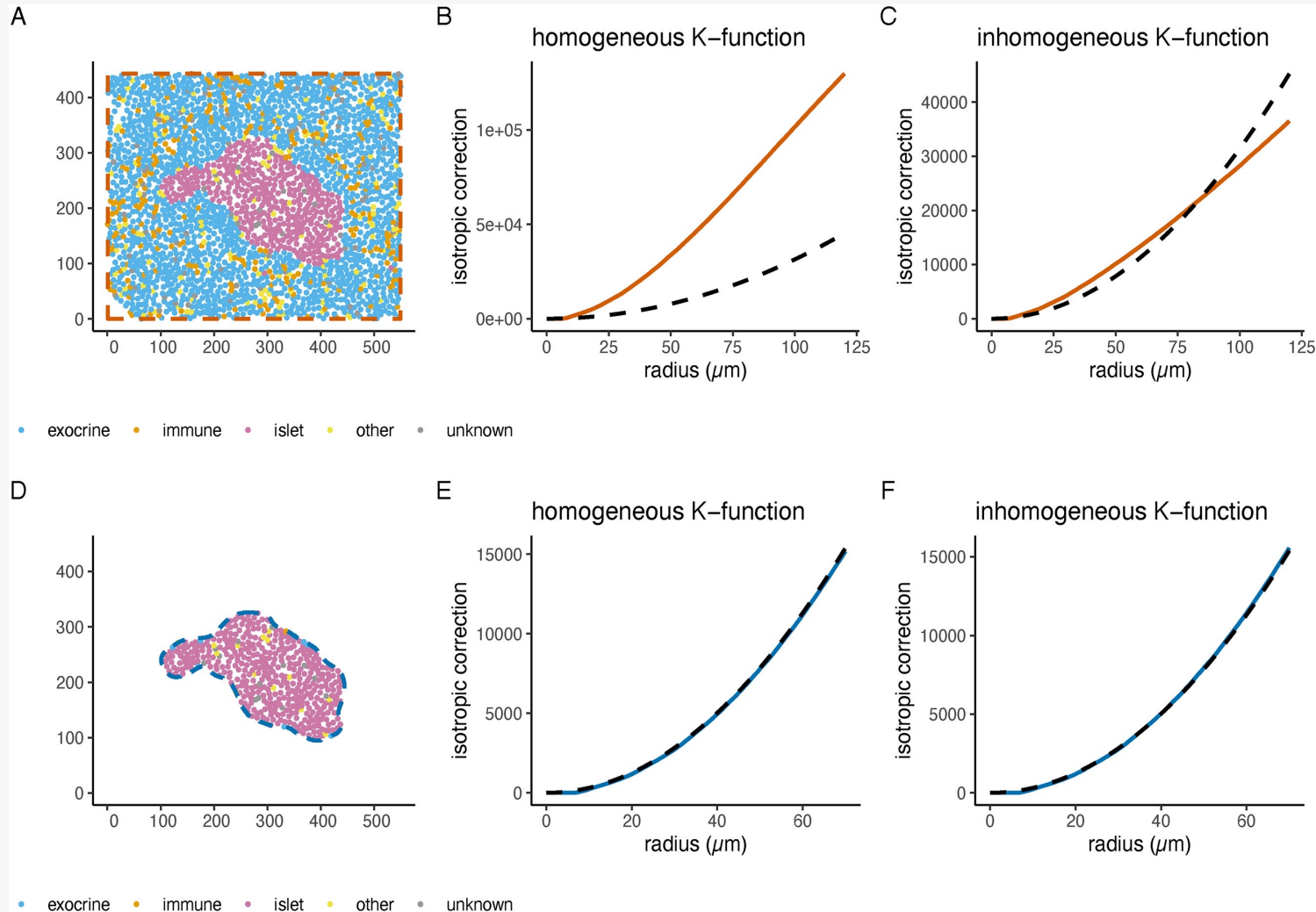


- E
- Whether you assume homogeneity or not (in the K-function calculation) can have a big impact on the estimated curves.
 - Are these cells clustered or is this due to varying intensity?





Observation window and spatial scale matter

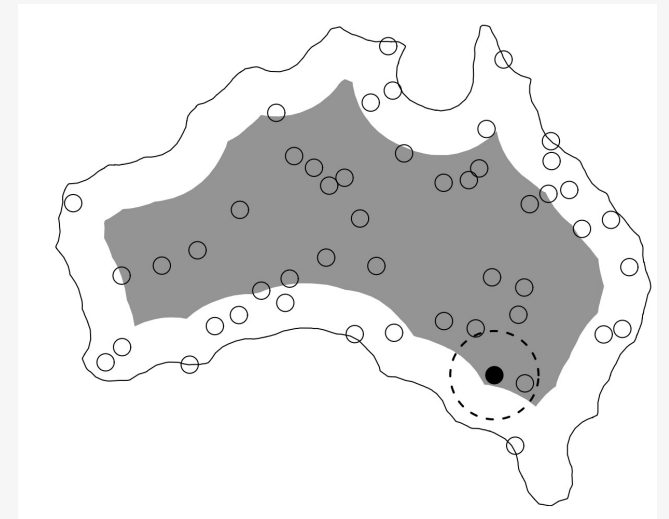


- Whole field of view (FOV): strong apparent clustering
- Restricted window: little evidence of clustering
- Different windows can lead to different spatial interpretations.

Edge effects

Points near the boundary have part of their neighbourhood outside the observed window.

→ naive distance summaries are biased near edges



Border correction



Extension of the K function (1): L and g functions

- $K(r)$: cumulative interaction up to distance r $K_{\text{pois}}(r) = \pi r^2$

- $L(r)$: rescaled K , with CSR baseline $L(r) = r$
$$L(r) = \sqrt{\frac{K(r)}{\pi}}$$

- $g(r)$: local interaction around distance r , roughly the derivative of K

$$g(r) = \frac{K'(r)}{2\pi r}$$

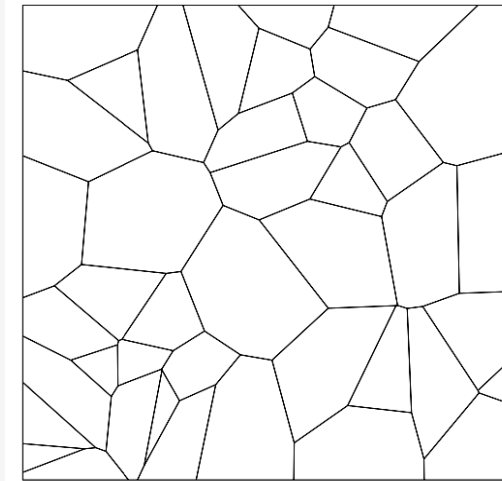
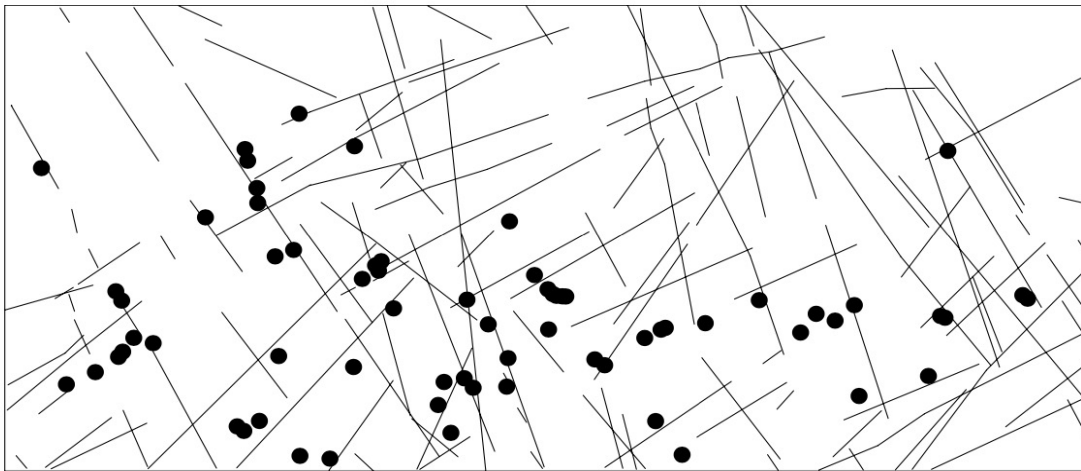


Point patterns can carry additional information

Marks: attributes attached to points, e.g., cell type, gene expression

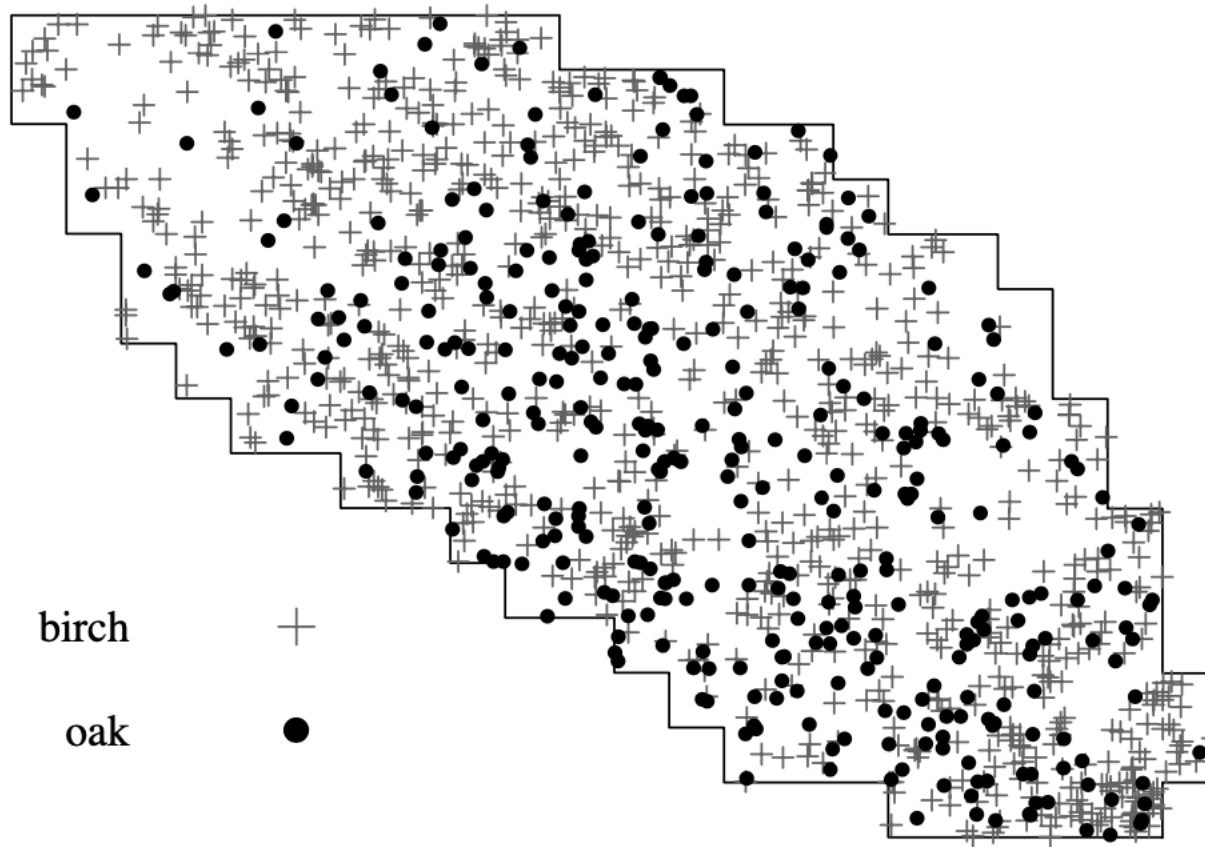
Types: categorical, continuous

Covariates: e.g., tissue region, distance to boundary





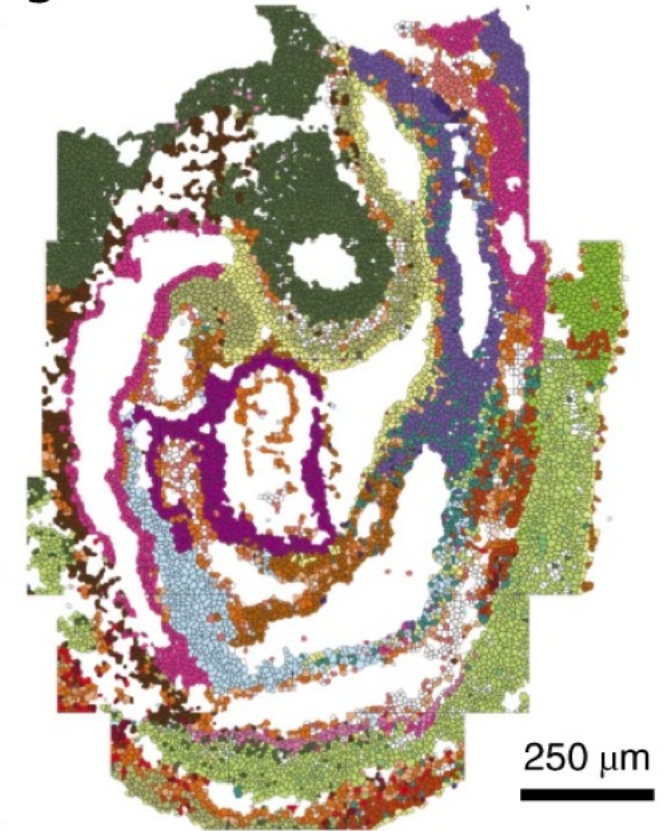
Multi-type point patterns



Baddeley et al., Spatial Point Patterns, 2015

b

- Allantois
- Anterior somitic tissues
- Blood progenitors
- Cardiomyocytes
- Caudal mesoderm
- Cranial mesoderm
- Definitive endoderm
- Dermomyotome
- Endothelium
- Erythroid
- ExE endoderm
- Forebrain/midbrain/hindbrain
- Gut tube
- Hematoendothelial progenitors
- Intermediate mesoderm
- Lateral plate mesoderm
- Mixed mesenchymal mesoderm
- Neural crest
- NMP
- Presomitic mesoderm
- Sclerotome
- Spinal cord
- Splanchnic mesoderm
- Surface ectoderm



Lohoff et al., Nat Biotechnol. 2022

Multi-type point patterns vs. multivariate point patterns



Extension of the K function (2): multi-type K

- Also called bivariate or cross-type K-function
- Expected number of type j points within distance r of a typical type i point, adjusted for the intensity of type j .



Other point pattern summaries

<u>Fest</u>	empty space function F
<u>Gest</u>	nearest neighbour distribution function G
<u>Jest</u>	J -function $J = (1 - G)/(1 - F)$
<u>Kest</u>	Ripley's K -function
<u>Lest</u>	Besag L -function
<u>Tstat</u>	Third order T -function
<u>allstats</u>	all four functions F, G, J, K
<u>pcf</u>	pair correlation function
<u>Kinhom</u>	K for inhomogeneous point patterns
<u>Linhom</u>	L for inhomogeneous point patterns
<u>pcfinhom</u>	pair correlation for inhomogeneous patterns
<u>Finhom</u>	F for inhomogeneous point patterns
<u>Ginhom</u>	G for inhomogeneous point patterns
<u>Jinhom</u>	J for inhomogeneous point patterns

<u>localI</u>	Getis-Franklin neighbourhood density function
<u>localK</u>	neighbourhood K -function
<u>localpcf</u>	local pair correlation function
<u>localKinhom</u>	local K for inhomogeneous point patterns
<u>localLinhom</u>	local L for inhomogeneous point patterns
<u>localpcfinhom</u>	local pair correlation for inhomogeneous patterns
<u>Ksector</u>	Directional K -function
<u>Kscaled</u>	locally scaled K -function
<u>Kest.fft</u>	fast K -function using FFT for large datasets
<u>Kmeasure</u>	reduced second moment measure
<u>envelope</u>	simulation envelopes for a summary function
<u>varblock</u>	variances and confidence intervals for a summary function
<u>lohboot</u>	bootstrap for a summary function



Point pattern summary

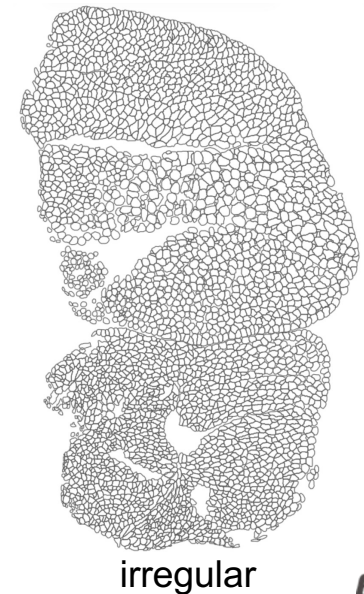
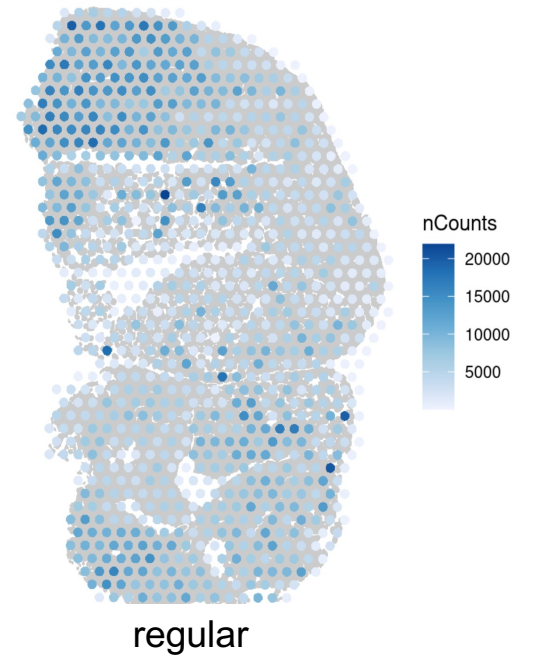
- ❑ First-order spatial structure → intensity, homogeneity
- ❑ Spatial interaction across scales → K , L , g
- ❑ Do different cell types co-localise? → cross- K / cross- L
- ❑ What assumptions matter? → observation window, edge effects, intensity assumption

Lattice data



What is lattice data?

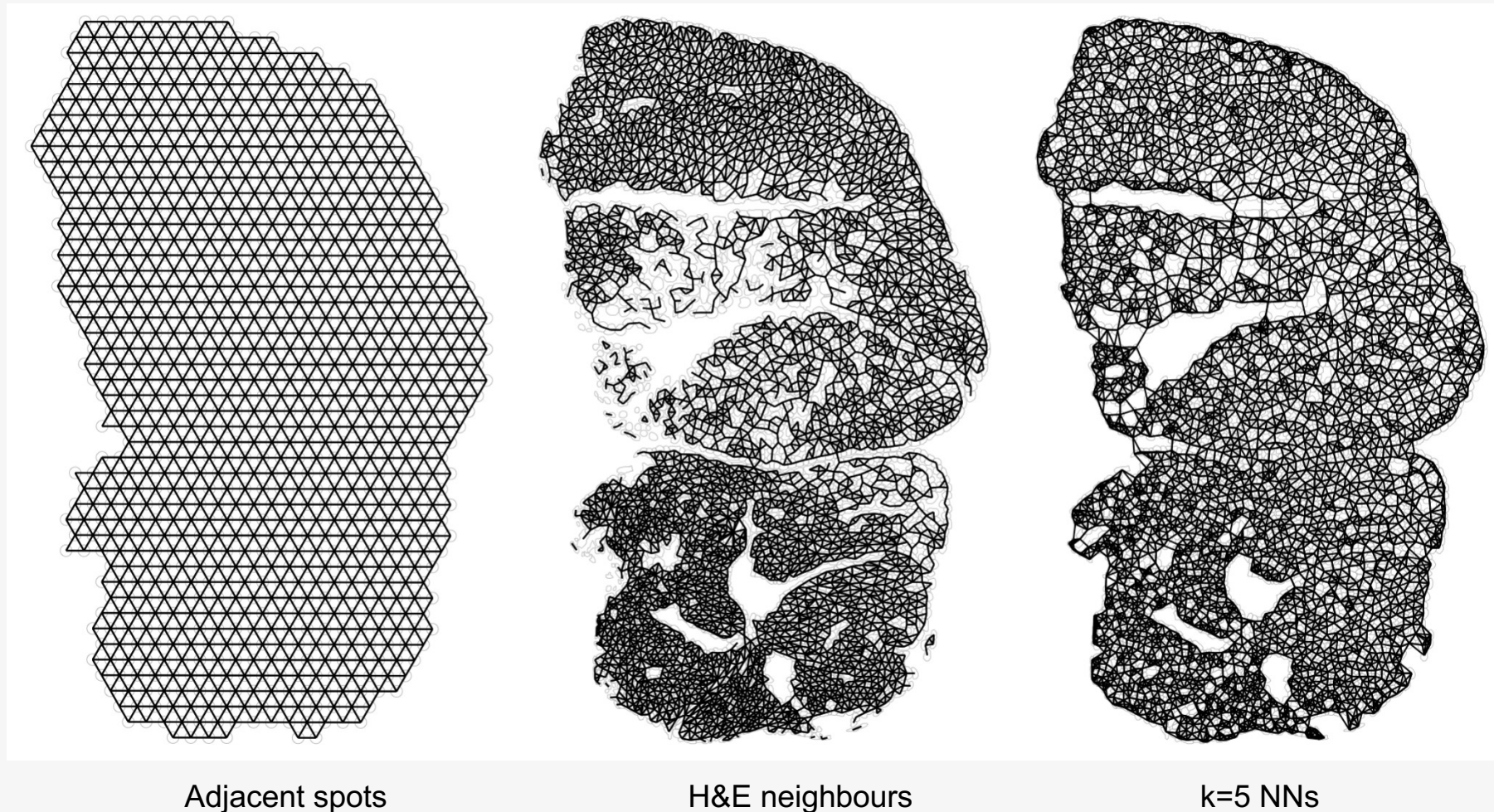
- Fixed spatial units
- Measurements are attached to those units
- Regular lattice: uniform units arranged on a regular grid
- Irregular lattice: variable shapes, sizes or spacing





Spatial weights matrix

- $W = (w_{ij})$ defines which spatial units are connected, and how strongly.





Global spatial autocorrelation

Global Moran's I

- Weighted association between a value and its spatial neighbours
- Positive: similar values tend to be close (high-high, low-low)

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Geary's C

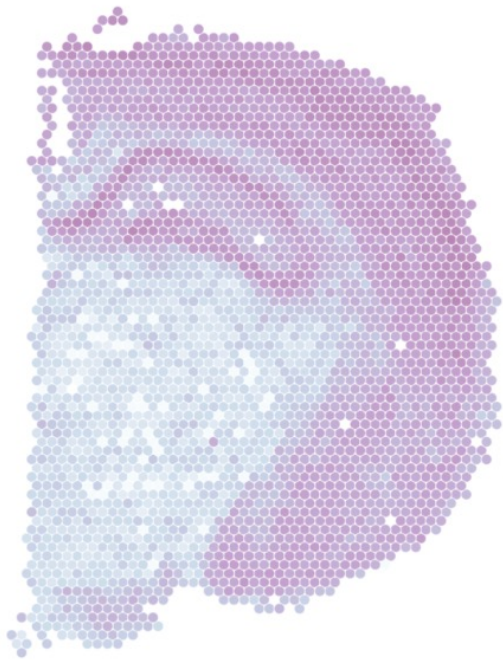
- Weighted squared differences between neighbours
- More sensitive to local differences than Moran's I

$$C = \frac{(n-1)}{2 \sum_{i=1}^n \sum_{j=1}^n w_{ij}} \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - x_j)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

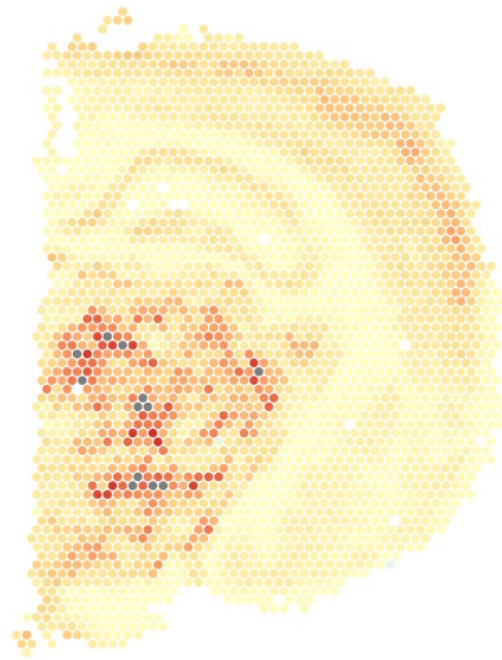


Local Moran's I

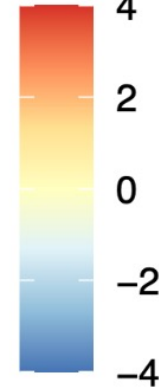
$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x}) (x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \longrightarrow I_i = (n-1) \frac{(x_i - \bar{x}) \sum_{j=1}^n w_{ij} (x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$



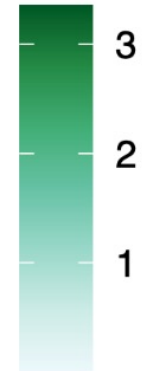
Nrgn



locl(Nrgn)



$-\log_{10}(\text{adj. p-val.})$



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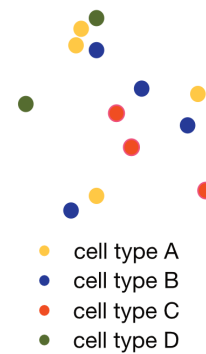
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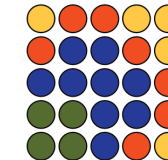
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Spatial autocorrelation of a gene?

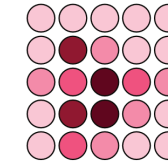
Moran's I and relatives

Spatial correlation of two genes?

Bivariate Moran's I and relatives

Spatial correlation of a set of genes?

Multivariate Geary's C



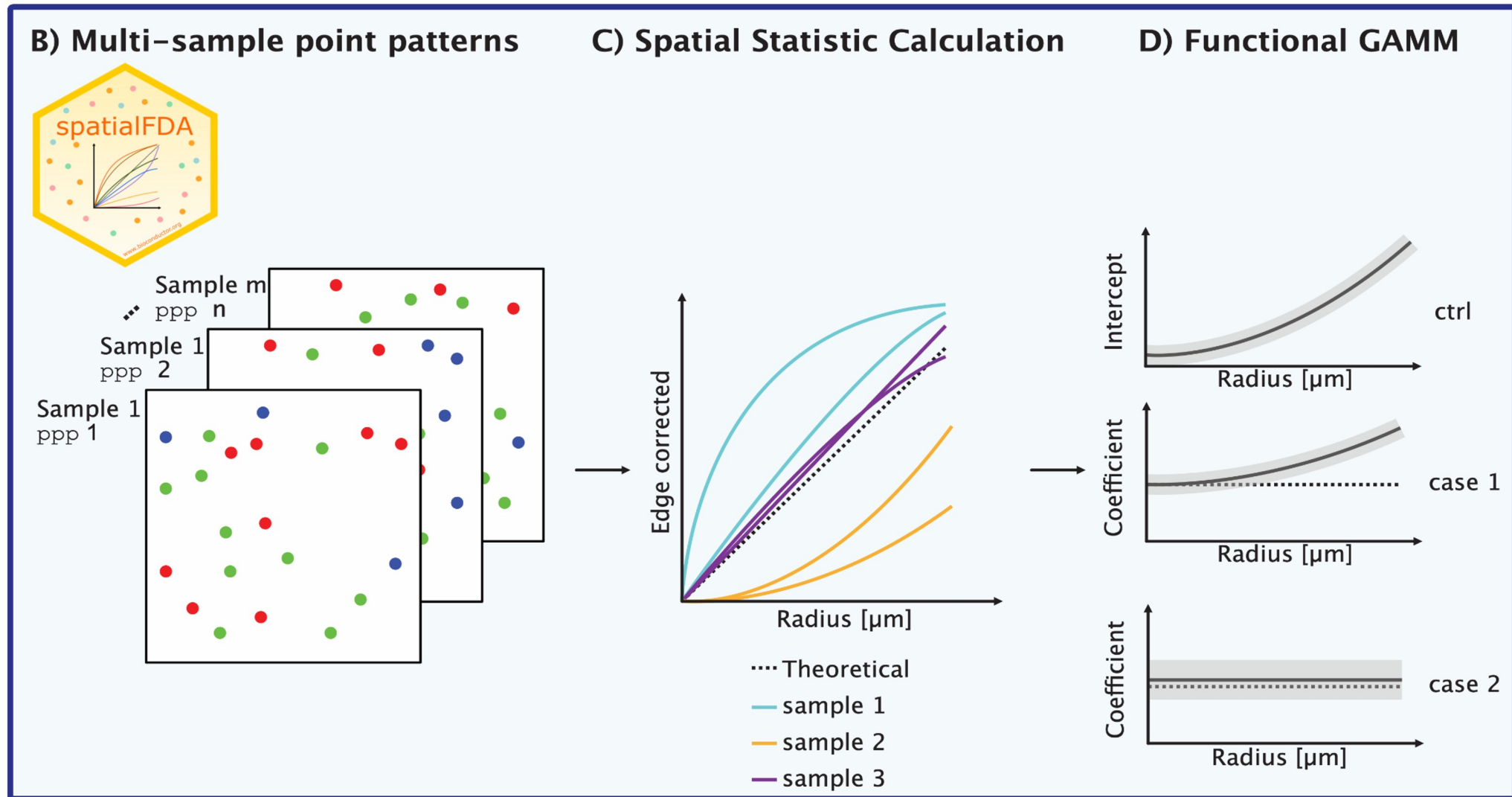
From a spatial statistic to multi-sample inference



spatialFDA: Differential co-localisation across scales



Martin





Functional generalised additive mixed model (fGAMM)

functional GAMM

$$\underbrace{\mathbb{E}[y_i(\boldsymbol{r})|X_i, g(i)]}_{\text{functions}} = \underbrace{g^{-1}}_{\text{link}} \left(\underbrace{\alpha(\boldsymbol{r})}_{\text{intercept}} + \underbrace{\beta_{0,g(i)}(\boldsymbol{r})}_{\text{random intercept}} + \underbrace{\sum_{j=1}^J f_j(X_{ji}, \boldsymbol{r})}_{\text{predictors}} \right)$$

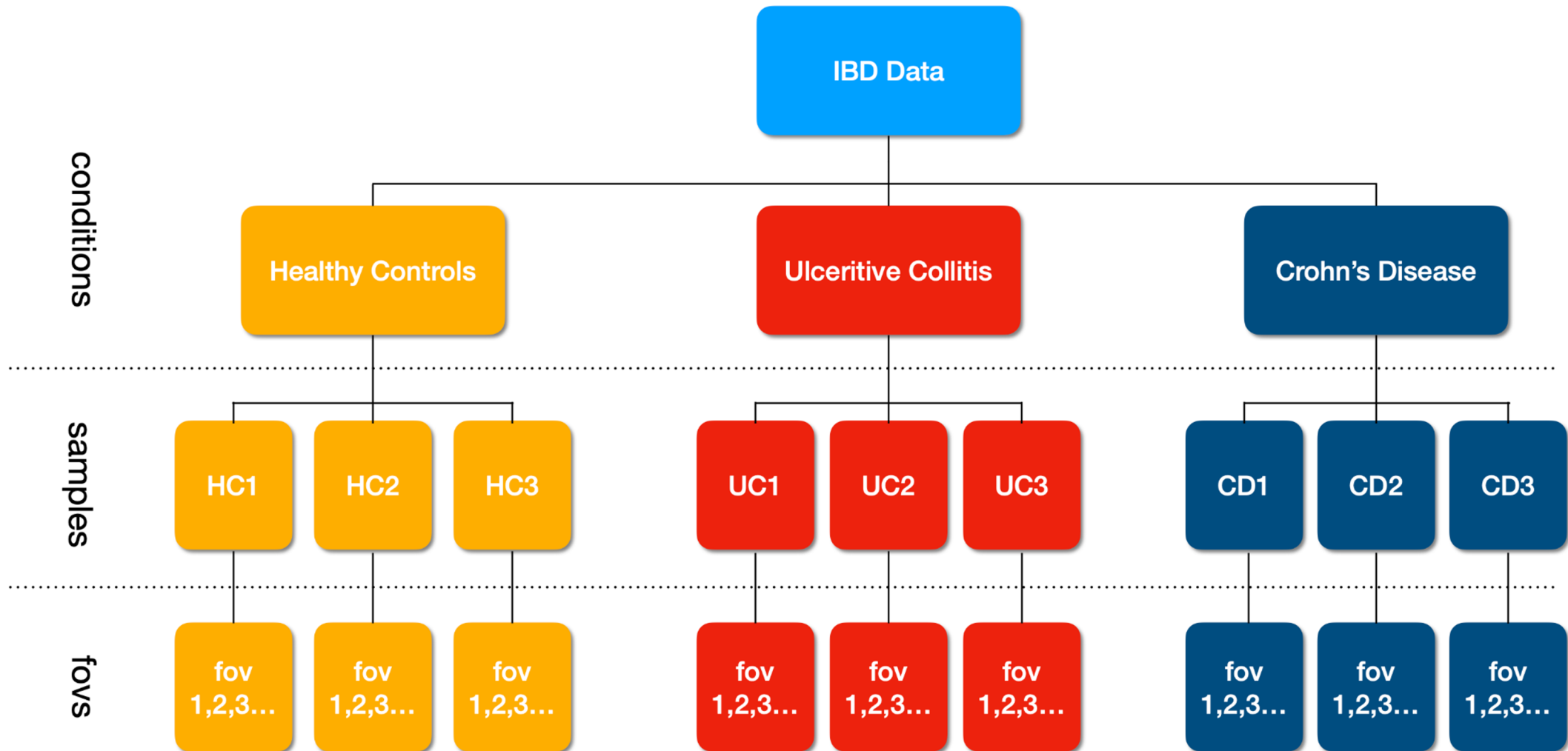
F-test of overall effect

$$H_0 : \beta(\boldsymbol{r}) = 0, \forall \boldsymbol{r} \in [r_{min}, r_{max}]$$

$$H_1 : \exists \boldsymbol{r} \in [r_{min}, r_{max}] \text{ s.t. } \beta(\boldsymbol{r}) \neq 0$$

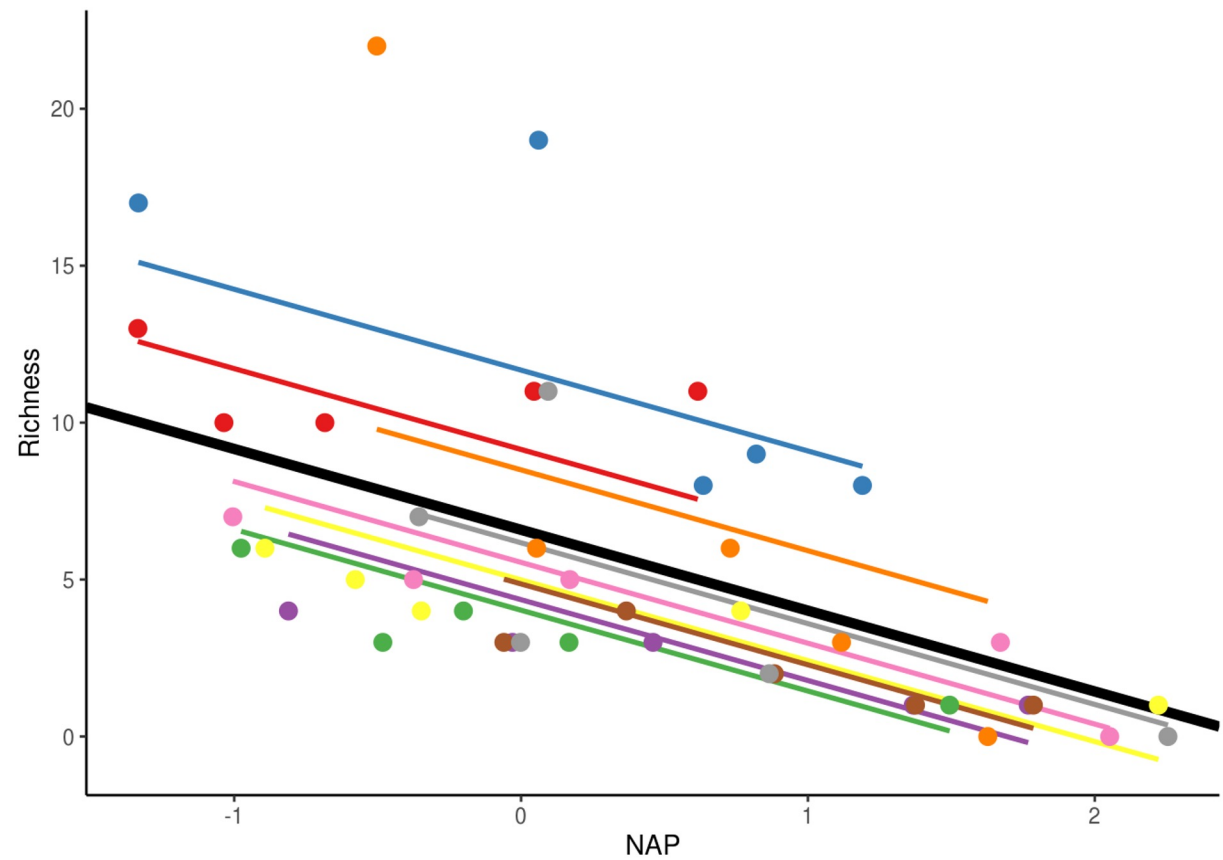
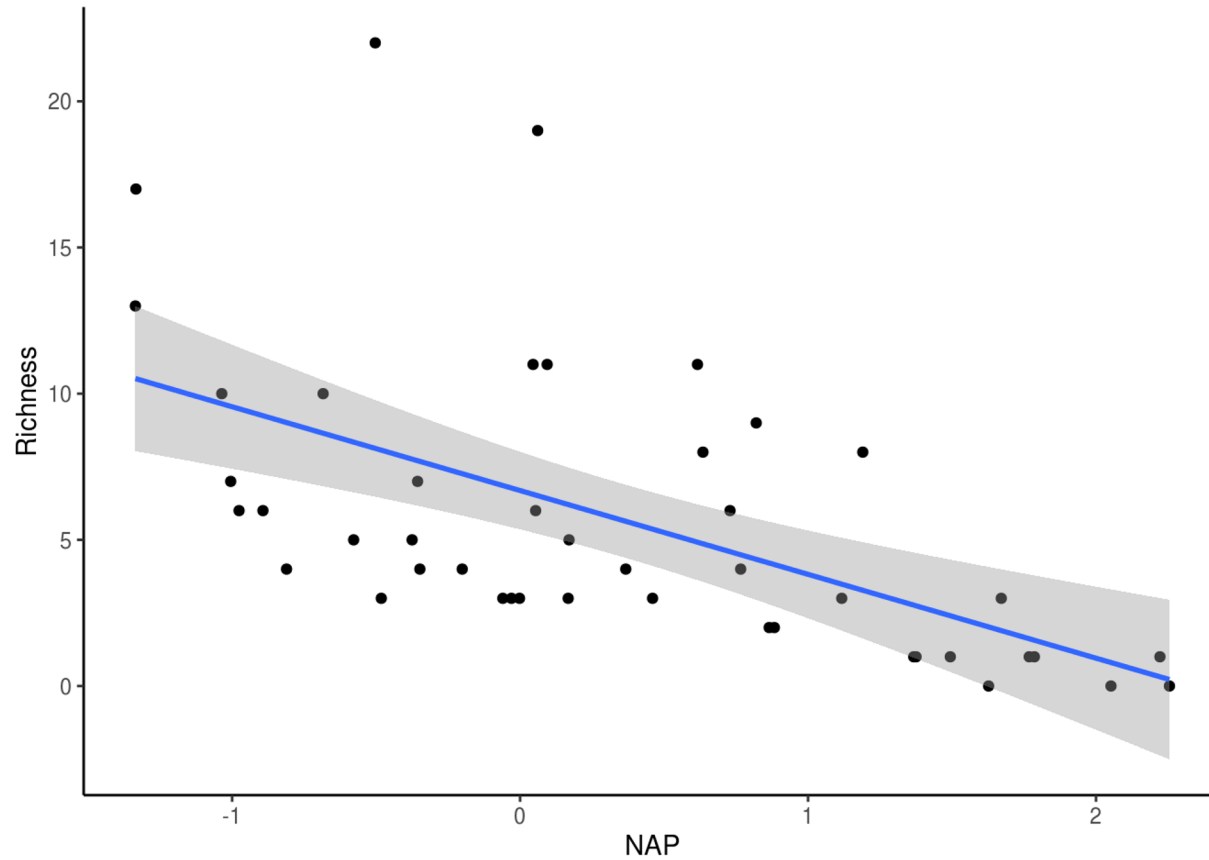


Modern spatial datasets show nested covariances





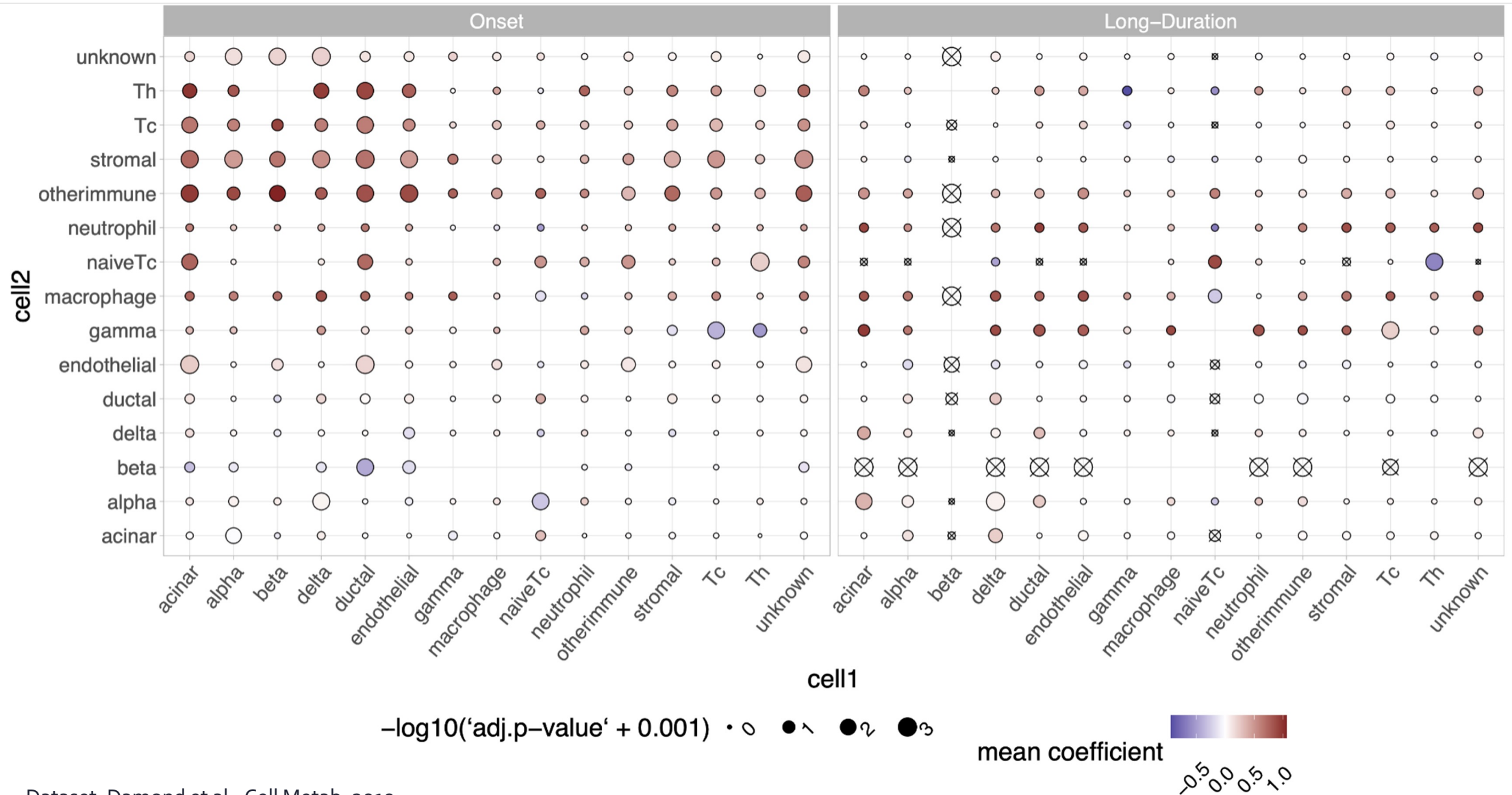
Random intercepts account for group effects



adapted from [Santangelo, 2018], licensed as CC BY 4.0



Overall F -test results of pairwise cell type combinations



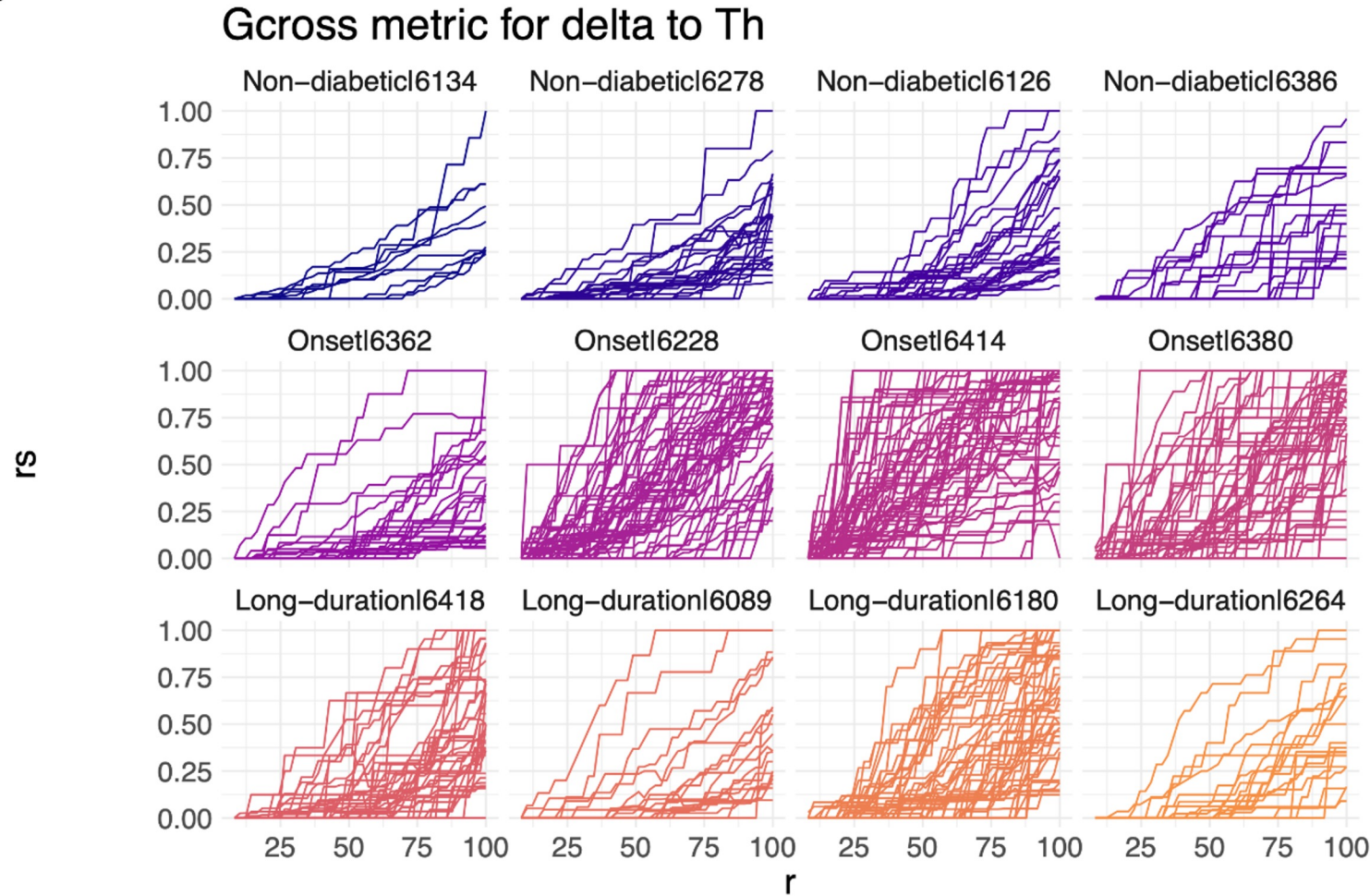
Dataset: Damond et al., Cell Metab. 2019

Emons et al., bioRxiv 2026



fGAMM shows scale-effect of delta-Th interaction

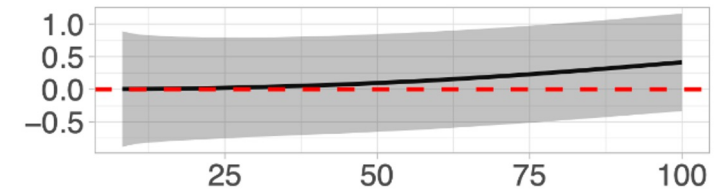
B



Dataset: Damond et al., Cell Metab. 2019

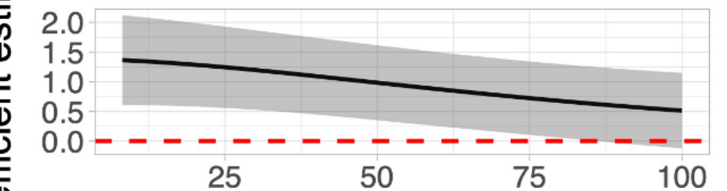
C

Intercept



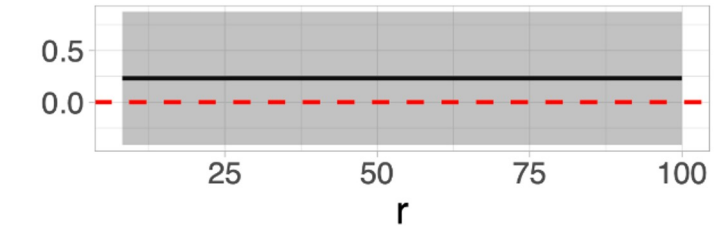
D

Onset



E

Long-Duration



Analysis of multicellular anatomical structures

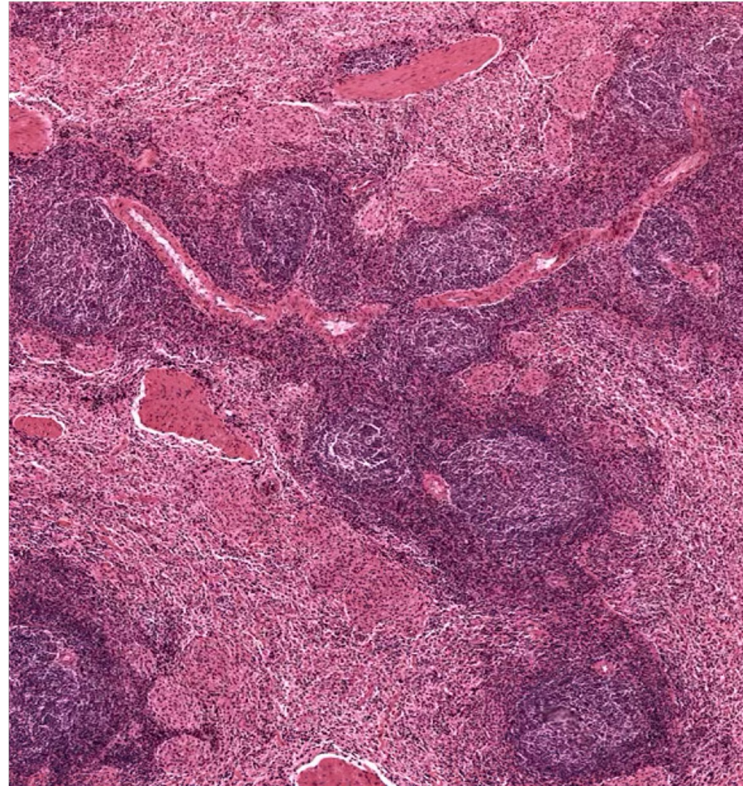


Biological function is often reflected in morphology and location

Ileum: villi and crypt



Spleen: red and white pulp



Ureter



Virginia Commonwealth University - Digital Histology; Pakurar, Bigbee *et al.*; <https://digitalhistology.org/>; (CC BY-NC-SA 4.0)

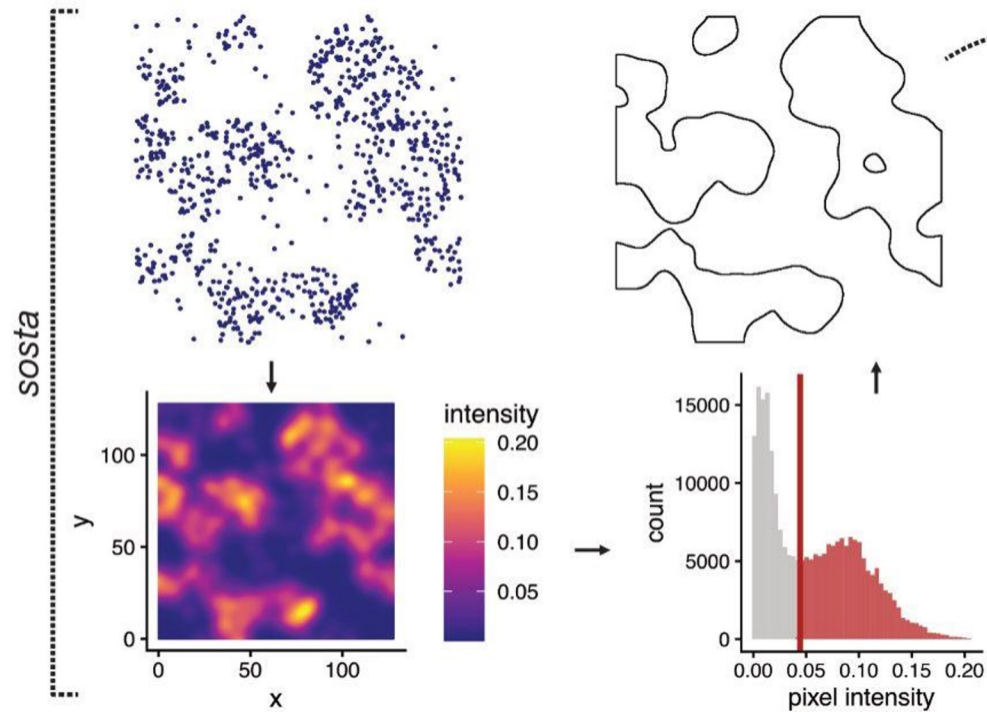


Analysis of anatomical structures using *sosta*

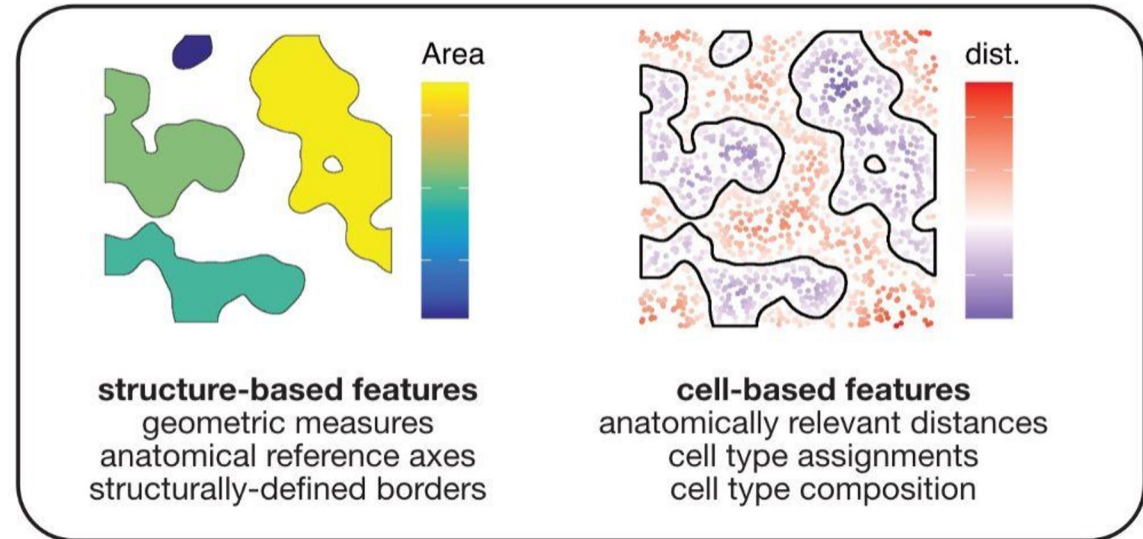


Samuel

Intensity-based segmentation of a point pattern

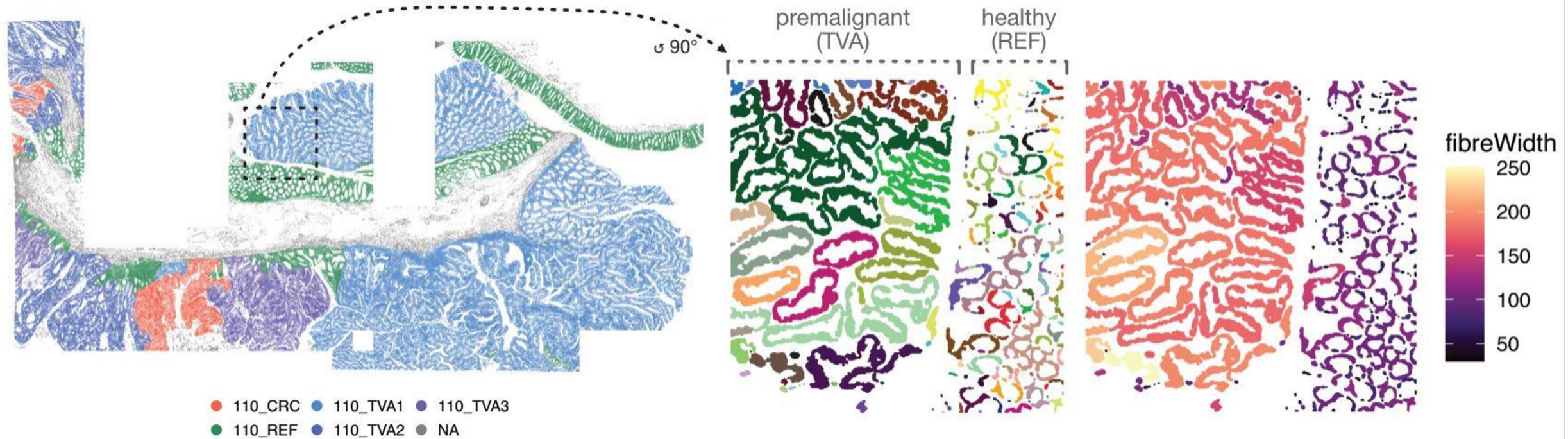


anatomical structure-based analysis





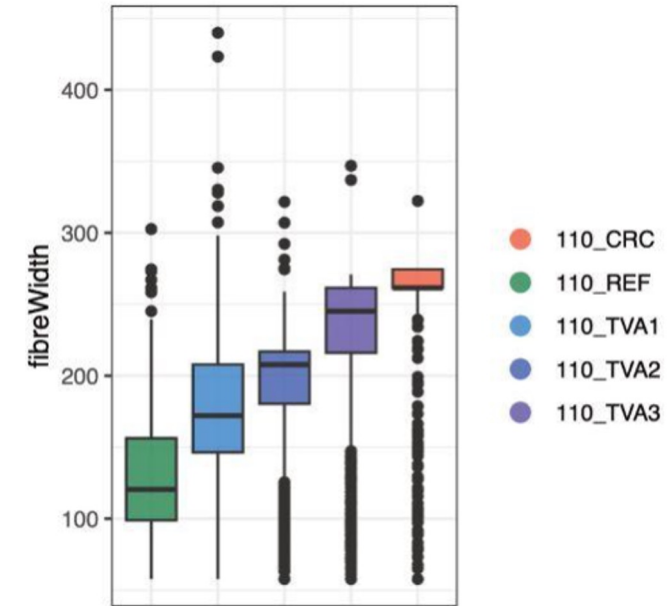
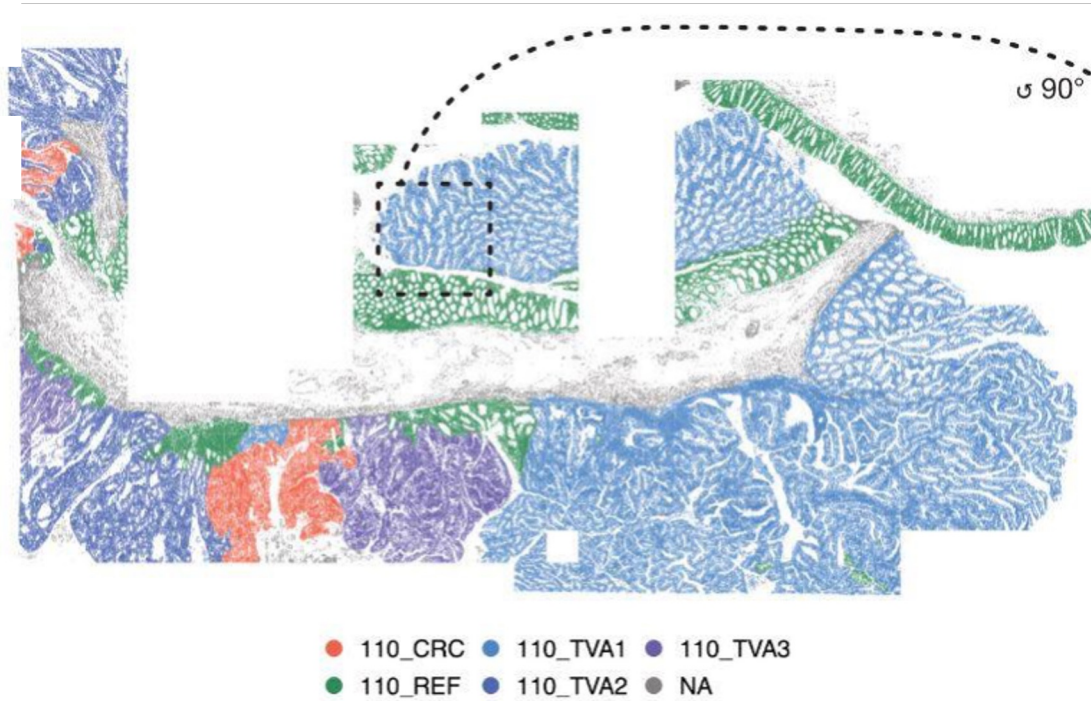
Structure-based analysis: Geometric features



Dataset: Crowell et al., bioRxiv 2025



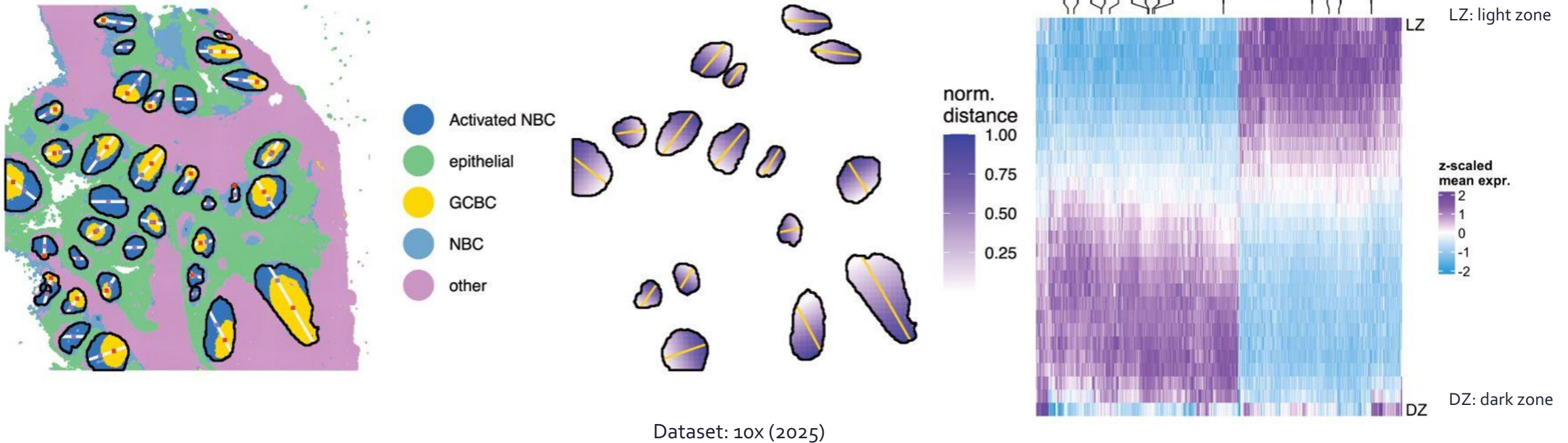
Structure-based analysis: Geometric features



- Fibre width increases along the pathological progression: healthy (HC) to tubulovillous adenoma (TVA) to colorectal cancer (CRC)
- Reflects gradual morphological changes related to increased crypt thickness.



Structure-based analysis: Gene expression



- Germinal centres are well defined anatomical structures with clear orientation
- Structure-based gene expression analysis to recover DZ / LZ markers



Other works in this space

SpaceANOVA: Spatial Co-occurrence Analysis of Cell Types in Multiplex Imaging Data Using Point Process and Functional ANOVA

Souvik Seal,^{*} Brian Neelon, Peggi M. Angel, Elizabeth C. O'Quinn, Elizabeth Hill, Thao Vu, Debashis Ghosh, Anand S. Mehta, Kristin Wallace, and Alexander V. A.

Bioimage informatics

spicyR: spatial analysis of *in situ* cytometry data in R

Nicolas P. Canete ^{1,2}, Sourish S. Iyengar³, John T. Ormerod³, Heeva Baharlou ^{1,2}, Andrew N. Harman^{1,2} and Ellis Patrick ^{1,3,*}

¹The Westmead Institute for Medical Research, University of Sydney, Westmead, NSW, Australia, ²Sydney Medical School, University of Sydney, Sydney, NSW, Australia and ³School of Mathematics and Statistics, University of Sydney, Sydney, NSW, Australia

Posted January 22, 2025.

STExplorer: Navigating the Micro-Geography of Spatial Omics Data

 Eleftherios Zormpas,  Nikolaos I. Vlachogiannis,  Anastasia Resteu,  Adrienne Unsworth,  Simon Tual-Chalot,  Birthe Dorgau,  Rachel Queen,  Majlinda Lako,  Dina Tiniakos,  Alexis Comber,  Quentin M. Anstee,  Antonis Giakountis,  Simon J. Cockell

doi: <https://doi.org/10.1101/2025.01.17.633539>

RESEARCH ARTICLE

FunSpace: A functional and spatial analytic approach to cell imaging data using entropy measures

Thao Vu ^{*}, Souvik Seal, Tusharkanti Ghosh, Mansooreh Ahmadian , Julia Wrobel[¶], Debashis Ghosh



Take-home messages

Data representation

Point patterns: locations are the outcome

Lattice data: values are measured at fixed locations

Spatial scale matters

Radius, neighbourhoods and spatial weights define what “local” means

Assumptions matter

Observation window, intensity assumptions and edge correction can change the interpretation

From description to inference

Spatial summaries and anatomical structures can be compared across samples, conditions



Appendix: resources

Spatial statistics book

- ❖ Baddeley, Rubak & Turner, [*Spatial Point Patterns*](#)
- ❖ Pebesma & Bivand, [*Spatial Data Science*](#)
- ❖ Holmes & Huber, [*Modern Statistics for Modern Biology*](#)

Related tutorials & packages

- ❖ [Pasta](#): Point-pattern and lattice methods, with R/Python
- ❖ OSTA: [Spatial statistics](#), [Differential colocalization](#), [Structure-based analysis](#) chapters
- ❖ [Voyager](#): lattice/geospatial analysis
- ❖ [spatialFDA](#): multi-sample comparison of spatial summary functions
- ❖ [sosta](#): structure-based spatial omics analysis



Appendix: references

- Baddeley, Adrian. "Modeling Strategies." In *Handbook of Spatial Statistics*, pp. 339–369. Chapman & Hall/CRC, 2010.
- Baddeley, Adrian., Rubak, Ege, and Turner, Rolf. "Spatial Point Patterns: Methodology and Applications with R (1st ed.)." Chapman and Hall/CRC. 2015.
- Canete, Nicolas P., et al. "spicyR: spatial analysis of in situ cytometry data in R." *Bioinformatics* 38.11 (2022): 3099–3105.
- Crowell, Helena L., et al. "Tracing colorectal malignancy transformation from cell to tissue scale." *bioRxiv* (2025): 2025.06.23.660674.
- Damond, Nicolas, et al. "A map of human type 1 diabetes progression by imaging mass cytometry." *Cell Metabolism* 29.3 (2019): 755–768.e5.
- Emons, Martin, et al. "Harnessing the potential of spatial statistics for spatial omics data with pasta." *Nucleic Acids Research* 53.17 (2025): gkaf870.
- Emons, Martin, et al. "Differential co-localisation analysis of multi-sample and multi-condition experiments with spatialFDA." *bioRxiv* (2026). doi:10.64898/2026.04.13.718197.
- Gunz, Samuel, Helena L. Crowell, and Mark D. Robinson. "Analysis of multicellular anatomical structures from spatial omics data using sosta." *PLoS Computational Biology* 22.9 (2026): e1014551.
- Lohoff, T., et al. "Integration of spatial and single-cell transcriptomic data elucidates mouse organogenesis." *Nature Biotechnology* 40.1 (2022): 74–85.
- Santangelo, James S. "Linear Mixed-Effects Models." *UofT Coders*, 2018, uoftcoders.github.io/course/leco8-linear-mixed-effects-models.html. Accessed 8 Dec. 2025.
- Seal, Souvik, et al. "SpaceANOVA: Spatial Co-occurrence Analysis of Cell Types in Multiplex Imaging Data Using Point Process and Functional ANOVA." *Journal of Proteome Research* 23.4 (2024): 1131–1143.
- Vu, Thao, et al. "FunSpace: A functional and spatial analytic approach to cell imaging data using entropy measures." *PLoS Computational Biology* 19.9 (2023): e1011490.
- Zormpas, Eleftherios, et al. "STExplorer: Navigating the Micro-Geography of Spatial Omics Data." *bioRxiv* (2025): 2025.01.17.633539.
- Virginia Commonwealth University. *Digital Histology*. Project Directors: Alice S. Pakurar and John W. Bigbee. CC BY-NC-SA 4.0.
- 10x Genomics. "Visium HD Spatial Gene Expression Library, Human Tonsil (Fresh Frozen)." Dataset, 2025.